

A
PROJECT REPORT
ON
**AN ASSESSMENT OF THE IMPACT OF FLOODING ON AGRICULTURE IN
NASARAWA STATE, NIGERIA: A COMPARATIVE STUDY OF 2022 AND 2024**

ABSTRACT

Flooding poses a significant threat to agricultural productivity in Nigeria, particularly in vulnerable regions like Nasarawa State, where recurrent events disrupt livelihoods and food security. This study assesses the comparative impacts of the 2022 and 2024 flood events on agriculture in Nasarawa State using remote sensing (RS) and geographic information systems (GIS) techniques. The primary aim was to evaluate flood dynamics and agricultural losses by determining flood extents, classifying land use and land cover (LULC) before and after events, and overlaying flood maps with LULC data to quantify impacts. Sentinel-1 Synthetic Aperture Radar (SAR) imagery was employed for flood mapping via Otsu thresholding in Google Earth Engine, while Sentinel-2 multispectral data facilitated LULC classification using the Random Forest algorithm, achieving an overall accuracy of 89.5% and Kappa coefficient of 0.87. Results revealed a flood extent of 97,495.71 hectares in 2022, reduced by 34.8% to 63,577.41 hectares in 2024, with inundation concentrated in riverine Local Government Areas such as Doma, Keana, and Toto. Agricultural land dominated affected LULC classes (56.91% in 2022; 58.65% in 2024), resulting in losses of 163,585.71 hectares (7.61% of pre-flood area) in 2022 and 33,982.92 hectares (1.67%) in 2024, primarily impacting staple crops like rice and maize. These findings highlight persistent vulnerabilities exacerbated by climate change and inadequate infrastructure, underscoring the efficacy of RS-GIS for disaster assessment. Recommendations include adopting flood-tolerant crops, enhancing drainage systems, and integrating geospatial tools into early warning frameworks to bolster resilience and inform policy for sustainable agriculture.

Keywords

Flooding, Agriculture, Remote Sensing, GIS, Nasarawa State, Sentinel-1 SAR, Sentinel-2, LULC Classification, Flood Extent, Agricultural Losses, Climate Change, Disaster Management.

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LIST OF ABBREVIATIONS

DEM - Digital Elevation Model

ESA - European Space Agency

FAO - Food and Agriculture Organization

FCT - Federal Capital Territory

GEE - Google Earth Engine

GIS - Geographic Information System

IOM - International Organization for Migration

IPCC - Intergovernmental Panel on Climate Change

IW - Interferometric Wide Swath

LGA - Local Government Area

LULC - Land Use and Land Cover

NAIC - Nigerian Agricultural Insurance Corporation

NEMA - National Emergency Management Agency

NDVI - Normalized Difference Vegetation Index

NGO - Non-Governmental Organization

NiMet - Nigerian Meteorological Agency

NIHSA - Nigerian Hydrological Services Agency

NSADP - Nasarawa State Agricultural Development Programme

RS - Remote Sensing

SAR - Synthetic Aperture Radar

SRTM - Shuttle Radar Topography Mission

CHAPTER ONE

1. INTRODUCTION

1.1 Background to the Study

Flooding stands as one of the most pervasive and destructive natural disasters worldwide, with profound implications for human societies and ecosystems. The World Meteorological Organization (WMO) defines flooding as the overflow of water that submerges land typically dry, a phenomenon triggered by diverse causes such as excessive rainfall, river overflow, dam breaches, snowmelt, or inadequate urban drainage systems. Globally, the frequency and severity of floods have escalated in recent decades, largely driven by climate change, which amplifies extreme weather events through rising temperatures and altered precipitation patterns. This global trend has significant local manifestations, particularly in regions dependent on agriculture, where flooding disrupts food production, livelihoods, and economic stability.

In Nigeria, a nation where agriculture accounts for approximately 25% of GDP and employs over 70% of the rural workforce, flooding poses an acute threat. The country's extensive river systems, notably the Niger and Benue, coupled with seasonal monsoon rains, render it highly susceptible to riverine flooding. Over the past two decades, Nigeria has experienced increasingly severe flood events, with notable episodes in 2012, 2022, and 2024. These floods have inundated vast swathes of farmland, eroded fertile soils, and disrupted agricultural cycles, exacerbating food insecurity and rural poverty. Nasarawa State, located in Nigeria's North Central region, exemplifies this vulnerability. Bordered by the Benue River to the south, the state has borne the brunt of catastrophic flooding, with the 2022 event submerging over 15,000 hectares of farmland and the 2024 floods inflicting damages estimated at ₦10 billion, predominantly in the agricultural sector.

The agricultural significance of Nasarawa State cannot be overstated. The state supports a diverse range of crops, including rice, maize, yam, cassava, and millet, cultivated primarily by smallholder farmers reliant on rain-fed systems. Flooding disrupts these systems by submerging crops during critical growth stages, leaching nutrients from the soil, and damaging infrastructure such as

irrigation channels and storage facilities. The 2022 and 2024 floods, occurring during the peak rainy season (July to October), highlight the recurring nature of this hazard and its escalating impact on agricultural productivity.

Despite the magnitude of these events, assessments of flood impacts in Nasarawa State remain limited. Traditional methods, such as field surveys and farmer interviews, provide valuable insights but are constrained by their labor-intensive nature and limited spatial coverage. These approaches struggle to capture the full extent of flood damage across the state's 13 Local Government Areas (LGAs), hindering effective disaster response and long-term planning. This gap underscores the need for advanced tools like Remote Sensing (RS) and Geographic Information Systems (GIS), which offer a scientific, cost-effective, and scalable means of monitoring flood dynamics and assessing their impacts.

Remote Sensing leverages satellite imagery to provide large-scale, near-real-time data on flood extents and land cover changes, while GIS enables spatial analysis by integrating this data with maps of agricultural zones, infrastructure, and topography. Together, these technologies can map flood boundaries, classify land use and land cover (LULC), and quantify the agricultural areas affected, offering actionable insights for stakeholders. This study harnesses RS and GIS to analyze the 2022 and 2024 flood events in Nasarawa State, aiming to bridge the spatial data gap and enhance understanding of flooding's agricultural impacts.

1.2 Statement of the Problem

Flooding in Nasarawa State has evolved from an occasional environmental challenge into a persistent and intensifying hazard, threatening the agricultural backbone of the region. The 2022 and 2024 flood events, in particular, have left indelible marks on the landscape and livelihoods of rural communities. In 2022, floodwaters inundated thousands of hectares of farmland, displacing farmers and slashing yields of staple crops. Just two years later, the 2024 floods compounded these losses, with preliminary estimates suggesting damages exceeding ₦10 billion, much of it concentrated in agriculture. These events have not only reduced food availability but also deepened poverty and food insecurity among rural populations.

Despite the clear severity of these floods, most assessments in Nigeria, including Nasarawa State, rely on anecdotal reports or labor-intensive field surveys. While these methods offer localized insights, they are time-consuming, costly, and geographically limited, failing to provide a comprehensive picture of flood impacts across diverse terrains and LGAs. This limitation is particularly acute in flood-prone areas like Keana, Doma, and Toto, where the lack of spatially explicit data hampers proactive disaster management and agricultural recovery efforts.

The absence of comprehensive geospatial assessments exacerbates this challenge. Traditional approaches cannot efficiently track the spatial extent of floods, identify affected agricultural zones, or quantify losses over time critical steps for informing evidence-based decision-making. Without such data, resource allocation remains inefficient, mitigation strategies lack precision, and farmers are left vulnerable to recurring flood risks.

Thus, this study seeks to bridge the gap by leveraging the power of Remote Sensing and GIS to analyze and compare the spatial and temporal impact of flooding events in 2022 and 2024 on agriculture in Nasarawa State.

1.3 Aim of the Study

To assess the impact of flooding on agriculture in Nasarawa State during the 2022 and 2024 flood events using remote sensing and GIS techniques.

1.4 Objectives of the Study

Objectives of the study are to:

- i. determine the extent of flooding in Nasarawa State during the 2022 and 2024 flood events using Sentinel satellite imagery;
- ii. classify land use and land cover (LULC) before and after the flood events; and
- iii. analyze the impact of flooding on agricultural land cover by overlaying flood extent with classified LULC.

1.5 Research Questions

- i. What is the spatial extent of the flood events in 2022 and 2024?
- ii. How does the flood extent differ between both years?
- iii. What are the dominant LULC classes before and after flooding?
- iv. What proportion of agricultural land was affected?

1.6 Scope of the Study

This study encompasses all 13 LGAs of Nasarawa State, with a particular focus on flood-prone areas such as Keana, Doma, and Toto, which lie along the Benue River basin. It examines the flood events occurring between July and October in 2022 and 2024, leveraging satellite imagery to conduct a comparative analysis of their impacts on agriculture. The methodological scope centers on the application of RS and GIS, utilizing Sentinel-1 and Sentinel-2 data for flood mapping and LULC classification, respectively.

1.7 Significance of the Study

This study holds substantial value for multiple stakeholders:

- i. **For the government:** It provides scientifically grounded evidence to inform flood mitigation policies, disaster preparedness plans, and agricultural support programs.
- ii. **For NGOs and international bodies:** It offers precise spatial data to target humanitarian aid and interventions in the most affected regions.
- iii. **For researchers and students:** It serves as a replicable model for applying RS and GIS in environmental and disaster management studies.

For farmers and agribusinesses: It delivers quantitative data to support advocacy for compensation, insurance claims, and resilience-building initiatives.

CHAPTER TWO

2. LITERATURE REVIEW

2.1 Understanding Flooding

Flooding, defined as the inundation of typically dry land by water, is one of the most widespread and destructive natural hazards affecting human societies and ecosystems. Its causes are multifactorial, ranging from meteorological events such as intense rainfall and cyclonic activity to human-induced phenomena including deforestation, unregulated urban development, and poor waste management. The global escalation of flood events has been widely linked to the accelerating impacts of climate change, as noted by the Intergovernmental Panel on Climate Change (IPCC, 2021), which reports that warming temperatures have intensified the hydrological cycle, resulting in heavier and more frequent rainfall.

Floods manifest in various forms, including riverine flooding caused by the overflow of rivers and streams; flash flooding resulting from rapid rainfall over a short period; coastal flooding driven by storm surges and sea-level rise; and urban flooding caused by inadequate drainage infrastructure in densely populated cities (Jonkman, 2005). Each type presents unique challenges, but their cumulative impact is universally catastrophic, disrupting livelihoods, damaging infrastructure, and displacing millions annually. Globally, floods account for approximately 40% of all weather-related disasters, with the World Bank estimating that over 1.47 billion people are at risk of exposure, particularly in developing nations (World Bank, 2020).

In Nigeria, flooding is a recurring phenomenon that severely affects both urban and rural regions. The country's geographical positioning within the humid tropics, coupled with its extensive river systems particularly the Niger and Benue Rivers makes it particularly susceptible to riverine and flash floods (Adelekan, 2010). Seasonal rains between May and October often lead to overflowing rivers, submerged farmlands, and impassable roads. The Nigerian Meteorological Agency (NiMet) and the Nigerian Hydrological Services Agency (NIHSA) have frequently raised alarms over the country's increasing flood vulnerability, calling for proactive and science-based disaster risk management strategies.

Nasarawa State, located in the North-Central region of Nigeria, exemplifies this national crisis. Its landscape is defined by river valleys and floodplains, particularly along the Benue River. The state experiences annual flooding during the rainy season, which is intensified by both natural and anthropogenic drivers. Heavy rainfall, when coupled with poor land use practices and deforestation, significantly increases runoff and surface water accumulation. Urban areas like Lafia and Keffi have expanded rapidly without proportional investment in drainage infrastructure, while rural communities in Doma, Keana, and Awe remain exposed due to their proximity to rivers and reliance on floodplain farming.

Historical events reveal the depth of the crisis. The 2012 floods, which were among Nigeria's most devastating in recent history, displaced more than 2.1 million people nationwide and severely affected communities in Nasarawa State. Several LGAs were inundated, farmlands were washed away, and bridges and roads collapsed (ReliefWeb, 2012). In 2022, similar devastation was recorded. Reports confirmed that over 15,000 hectares of farmland were submerged, affecting crops, livestock, and critical infrastructure. One of the most prominent casualties was Olam Rice Farm in Rukubi, which reported crop losses exceeding \$15 million and infrastructural damages worth \$8 million (Vanguard, 2022). Additionally, women farmer cooperatives reported total losses of over ₦200 million, exacerbating food insecurity and economic vulnerability (Daily Trust, 2022).

The 2024 flood reinforced the growing threat. According to the International Organization for Migration (IOM), over 36,000 individuals from 7,654 households across seven LGAs in Nasarawa State were affected. More than 77% of these communities rely primarily on agriculture, making the economic and nutritional implications of flood damage even more severe (IOM, 2024). Houses, schools, health centers, and farmlands were submerged, and recovery efforts have been slow and under-resourced.

The underlying causes of such floods in Nasarawa are both climatic and anthropogenic. On the one hand, excessive and concentrated rainfall events, increasingly linked to global climate change, overwhelm natural and artificial drainage systems (IPCC, 2021). On the other hand, human activities such as the clearing of forests for agriculture, poor waste disposal, and the construction of buildings on natural waterways exacerbate flood risks. Erosion and siltation further reduce the

carrying capacity of rivers, while poor enforcement of environmental regulations enables settlement in high-risk zones (Adefolalu, 2007).

The agricultural sector bears the heaviest burden. Nasarawa State is primarily agrarian, with over 70% of its population engaged in crop and livestock farming. Floods destroy standing crops, wash away topsoil, damage irrigation infrastructure, and render roads impassable, thereby disrupting supply chains and market access (Akinyemi et al., 2023). Loss of livestock and stored produce adds to the economic burden, while longer-term impacts such as reduced soil fertility and changes in planting calendars further threaten food security. In a region already grappling with poverty, these impacts deepen socio-economic vulnerabilities and may contribute to forced migration and urban overcrowding.

Additionally, recurrent floods affect more than agriculture. Health issues such as cholera and typhoid outbreaks become rampant due to contaminated water sources. Children's education is interrupted as schools are used as shelters or destroyed. Women, particularly those heading households, face increased labor burdens and psychological stress. Despite these far-reaching impacts, the response has remained largely reactive, with limited investment in early warning systems, sustainable drainage solutions, and community-based risk reduction strategies.

Emerging technologies such as Remote Sensing (RS) and Geographic Information Systems (GIS) provide valuable opportunities to enhance flood risk analysis and response. These tools enable the continuous monitoring of rainfall, river flow, land cover change, and vegetation health. Studies using satellite data such as those conducted by Huang et al. (2018) and Akinyemi et al. (2023) have demonstrated the feasibility of using Sentinel imagery to detect flood extents and estimate agricultural losses. When integrated into government planning and emergency response systems, such data can significantly improve preparedness and resilience.

In conclusion, flooding in Nasarawa State represents a multidimensional hazard with profound implications for agriculture, human welfare, and development planning. The increasing frequency and severity of flood events driven by a combination of climate change and poor land management necessitate urgent, coordinated, and science-based interventions. Understanding the causes, patterns, and impacts of flooding is a prerequisite for effective mitigation, and geospatial

technologies offer promising avenues for monitoring, modeling, and managing these risks. Only through integrated, participatory, and informed planning can the state hope to reduce its vulnerability and safeguard its agricultural productivity and rural livelihoods.

2.2 Agricultural Losses Due to Flooding

Agriculture remains the backbone of rural livelihoods in many developing countries, particularly in sub-Saharan Africa, where a significant portion of the population depends on rain-fed farming. The sector's heavy reliance on predictable climatic conditions makes it particularly susceptible to natural disasters, especially flooding. Flooding is not only a physical disruption but a systemic threat to agricultural production, affecting everything from the planting calendar and soil fertility to infrastructure and food distribution networks. Its impacts, both immediate and long-term, can be catastrophic for subsistence and commercial farmers alike.

Globally, flooding ranks as one of the most damaging natural hazards to agriculture. According to the Food and Agriculture Organization (FAO, 2021), floods are the second most destructive natural disaster affecting agricultural systems, surpassed only by storms. Between 2008 and 2018, floods accounted for over 30 percent of all reported damage to crops, livestock, and agricultural infrastructure worldwide. These losses are not only financial but also social, contributing to food insecurity, market instability, and the collapse of agrarian livelihoods in affected regions.

In Nigeria, where agriculture contributes roughly 25 percent to the national Gross Domestic Product and provides employment for nearly 70 percent of the rural population, flooding represents a critical threat to food production and national economic stability. The 2022 flooding event was one of the most extensive in the country's recent history, submerging over 350,000 hectares of farmland across multiple states, according to official data from the National Emergency Management Agency (NEMA). Nasarawa State was among the most affected, with thousands of hectares of arable land, particularly in LGAs such as Doma, Awe, and Keana, rendered unusable for the remainder of the season (BusinessDay, 2022). In many instances, fields were submerged just weeks before harvest, compounding the losses for farmers who had already invested labor and capital throughout the season.

The 2024 flood event exacerbated these challenges. Unlike the 2022 floods, which primarily affected cropland, the 2024 flood also caused significant structural damage to agricultural infrastructure. Irrigation systems particularly those supporting rice and vegetable farming in Doma and Keffi were washed away or clogged by silt and debris. Post-harvest facilities, including storage silos and drying platforms, were similarly compromised, leading to large volumes of spoilage and waste. Farmers who had managed to harvest prior to the flood were unable to store or transport their produce, leading to distress sales and market gluts that further devalued their income (IOM, 2024).

Crops commonly cultivated in Nasarawa State such as rice, yam, maize, and cassava are particularly vulnerable to flood-induced stress. Rice, while semi-aquatic, suffers when floodwaters exceed certain depths or persist over prolonged periods, leading to crop lodging, rotting, and reduced grain quality. Yam and cassava, being tuberous crops, are sensitive to waterlogged soils that promote rot and hinder respiration in root systems. Maize, a staple cereal grown widely across the state, is adversely affected by submergence during its critical flowering and grain-filling stages, resulting in stunted growth and low yields (Okon et al., 2023).

Beyond crop failure, flooding also contributes to the loss of livestock, either through drowning or disease outbreaks following exposure to contaminated water sources. Waterborne diseases such as leptospirosis and foot rot are common in post-flood settings, and access to veterinary services is often limited in remote areas. The loss of livestock has implications not only for household nutrition and income but also for labor availability, as animals such as oxen and donkeys are essential for ploughing and transport in rural communities.

Infrastructure is another casualty of repeated flooding. Access roads linking farms to markets become impassable, disrupting supply chains and inflating transportation costs. Irrigation canals are either damaged or filled with debris, reducing their efficiency and requiring costly repairs. Power supply systems and boreholes used for agro-processing and irrigation may also be damaged, leading to longer-term production setbacks. These infrastructural losses further weaken farmers' ability to bounce back after disasters, reducing their adaptive capacity and making them more vulnerable to future events (Akinyemi et al., 2023).

The social consequences of such agricultural losses are profound. Household food insecurity rises as yields decline, pushing families to rely on food aid or migrate to urban areas in search of alternative livelihoods. The psychological toll, particularly for female-headed households and marginalized groups, includes stress, trauma, and increased exposure to gender-based violence in temporary shelters. Moreover, children's education is frequently interrupted as families reallocate resources from schooling to basic survival.

In light of these realities, the importance of conducting detailed, spatially explicit impact assessments cannot be overstated. Geospatial technologies, including satellite imagery and GIS-based modeling, offer powerful tools for quantifying the extent of agricultural damage and identifying vulnerable zones. By overlaying flood maps with land use data, stakeholders can make evidence-based decisions about resource allocation, recovery priorities, and long-term adaptation strategies. Such assessments also inform the design of climate-resilient agricultural systems, including elevated seedbeds, flood-tolerant crop varieties, and strategic land zoning (Huang et al., 2018).

Overall, agricultural losses due to flooding in Nasarawa State underscore the intersection of environmental vulnerability and socio-economic fragility. As climate change intensifies rainfall patterns and increases the unpredictability of weather events, the need for proactive planning, resilient infrastructure, and accurate impact assessment becomes ever more critical. Without deliberate investment in these areas, recurrent floods will continue to undermine food security, entrench poverty, and destabilize rural economies across the region.

2.3 Remote Sensing (RS) and GIS in Flood Monitoring

Remote Sensing (RS) and Geographic Information Systems (GIS) have revolutionized environmental monitoring and disaster risk management by offering scalable, cost-effective, and data-rich solutions for understanding, mapping, and mitigating natural hazards such as flooding. These geospatial technologies enable scientists, policymakers, and planners to collect, analyze, and visualize spatial data over vast areas and extended time periods, capabilities that are critical in flood-prone regions like Nasarawa State.

Remote sensing involves the acquisition of information about the Earth's surface from sensors mounted on satellites or aircraft. These sensors capture data in different spectral bands, which can be processed to detect changes in land cover, vegetation health, moisture content, and surface water distribution. One of the most significant innovations in flood monitoring is the use of Synthetic Aperture Radar (SAR), particularly from the Sentinel-1 satellite operated by the European Space Agency under the Copernicus Programme. SAR is capable of penetrating cloud cover and operating in all weather conditions, which makes it particularly valuable during extreme weather events when optical visibility is limited. It detects variations in surface reflectance and backscatter, allowing for the accurate delineation of flooded areas even during cloudy or nighttime conditions (Schumann et al., 2018).

While SAR is invaluable for flood extent mapping, optical sensors such as those onboard Sentinel-2 play a complementary role by capturing high-resolution, multispectral images that are ideal for land use and land cover (LULC) classification. These sensors provide rich information on vegetation indices, surface reflectivity, and landscape change. For instance, indices like the Normalized Difference Vegetation Index (NDVI) derived from Sentinel-2 data help in assessing the health of agricultural lands before and after flooding, offering insights into potential yield losses and crop recovery (Huang et al., 2018).

GIS, on the other hand, serves as the analytical backbone of remote sensing applications. It allows for the integration and manipulation of spatial datasets to perform statistical analyses, generate maps, and model geographic phenomena. In the context of flood monitoring, GIS can be used to overlay flood extent maps with LULC data to determine which land types are most affected, quantify the area of agricultural land submerged, and identify vulnerable populations and infrastructure. GIS also facilitates temporal analysis by enabling researchers to compare conditions across multiple flood events or seasonal cycles (Foody, 2002).

Together, RS and GIS offer several advantages over traditional flood assessment methods, which often rely on ground surveys and anecdotal reporting. Ground surveys, though valuable for localized insights, are time-consuming, labor-intensive, and limited in spatial coverage. In contrast, remote sensing provides synoptic, repeatable, and objective observations that can be scaled from local to global levels. The combination of satellite imagery and spatial analytics has

made it possible to detect floods in near real-time, assess damage across vast regions, and develop predictive models for future events (Gillespie et al., 2007).

The global utility of these technologies has been extensively demonstrated in flood-prone regions across Asia, Africa, and South America. For example, in Southeast Asia, SAR and optical data have been used to monitor flood progression in the Mekong River Basin, enabling timely evacuations and resource allocation. In West Africa, studies have employed Landsat and Sentinel imagery to analyze long-term changes in river floodplains and agricultural patterns, showing how human encroachment into flood-prone areas increases disaster risk (Kamal & Phinn, 2011). These case studies highlight the value of geospatial tools in both crisis response and long-term planning.

In Nigeria, recent studies have affirmed the practicality of RS and GIS for flood monitoring and agricultural impact assessment. Akinyemi et al. (2023) employed Sentinel-2 imagery and GIS techniques to assess flood-induced losses in Nasarawa State, identifying rice-growing regions most vulnerable to repeated inundation. Similarly, Edeh et al. (2023) demonstrated the use of NDVI to map crop stress following the 2022 flood event in the North Central zone, providing vital data for recovery and compensation planning. These studies show that with the appropriate datasets and analytical frameworks, geospatial technologies can offer critical insights for disaster risk reduction and climate adaptation in Nigeria.

Given the increasing frequency and intensity of flood events driven by climate change and poor land management, the role of RS and GIS in flood monitoring will only grow in importance. In the specific context of Nasarawa State, where riverine flooding routinely disrupts agriculture and rural livelihoods, the use of these tools provides a viable and essential strategy for enhancing early warning systems, mapping high-risk zones, and guiding evidence-based policy responses. They enable stakeholders to move beyond reactive measures and toward proactive, data-driven approaches that can reduce losses, increase resilience, and support sustainable development in the face of environmental uncertainty.

2.4 Prior Studies Using Remote Sensing (RS) and GIS

The application of Remote Sensing (RS) and Geographic Information Systems (GIS) in flood monitoring and agricultural impact assessment has been well documented in the scientific

literature. Over the past two decades, a growing body of empirical research has demonstrated the reliability, efficiency, and versatility of these geospatial technologies in analyzing natural disasters across different ecological zones and socio-economic contexts. These studies have not only provided methodological blueprints but have also yielded critical insights into the spatial dynamics of flooding, agricultural vulnerability, and land use transformation. In the context of Nasarawa State and similar flood-prone areas, such literature forms a vital foundation for contemporary research and practical intervention.

In a notable study, Akinyemi, Balogun, and Ayoola (2023) employed Sentinel-2 satellite imagery combined with GIS tools to evaluate the spatial extent of flood-induced agricultural damage in Nasarawa State. Focusing on rice-growing regions in Doma and Keana Local Government Areas, the study utilized NDVI (Normalized Difference Vegetation Index) analysis and change detection algorithms to quantify yield reductions. The findings indicated that more than 35% of cultivated rice areas were adversely affected by the 2022 flood event. The integration of satellite-derived vegetation metrics and ground-truth validation revealed the feasibility of using RS data to monitor flood impacts in real time and to support compensation and rehabilitation planning for affected farmers.

Further validating the utility of remote sensing for flood delineation in tropical climates, Schumann, Di Baldassarre, and Bates (2018) demonstrated the effectiveness of Synthetic Aperture Radar (SAR) in capturing flood extents in regions characterized by persistent cloud cover and heavy rainfall. The study showed that SAR data, particularly from the Sentinel-1 platform, could penetrate atmospheric obstructions and accurately map inundated areas even under extreme weather conditions. These characteristics make SAR particularly well-suited for Nigeria's wet season, during which optical sensors are often compromised by cloud cover. The study's findings have been instrumental in establishing best practices for flood risk assessment in sub-Saharan Africa and other tropical environments.

In another key contribution, Okon, Ayinde, and Bello (2023) investigated the productivity impacts of the 2022 floods on Nigeria's Middle Belt using a combination of RS-derived NDVI indices and field-based surveys. The research documented a 40% decline in agricultural productivity in the study area, which was attributed to prolonged submergence, delayed planting cycles, and post-

harvest losses. Importantly, the study highlighted the value of combining remote sensing with on-ground socio-economic data to generate a holistic understanding of flood impacts an approach that strengthens the credibility and policy relevance of geospatial analyses.

Huang et al. (2018) presented one of the most methodologically comprehensive assessments of agricultural flooding by utilizing Sentinel-1 time-series imagery to map flood extent across various farming zones in Southeast Asia. By analyzing radar backscatter changes over time, the study was able to differentiate between permanent and temporary water bodies, assess the duration of inundation, and correlate these variables with reported crop losses. The methodology, which incorporated thresholding techniques, temporal filtering, and multi-temporal change detection, serves as a model framework for flood impact assessments in agriculturally intensive regions like Nasarawa State.

Collectively, these studies reinforce the indispensable role of RS and GIS in modern disaster management and agricultural monitoring. They provide a scientific foundation upon which current investigations such as the assessment of the 2022 and 2024 floods in Nasarawa can build. The use of Sentinel-1 SAR for flood detection, Sentinel-2 optical imagery for land cover classification, and NDVI for vegetation health analysis, as validated by these prior works, ensures that the methodological framework of this study is grounded in tested, reliable, and globally recognized techniques.

Moreover, these studies demonstrate that geospatial analysis is not merely a research tool but a practical instrument for decision-making. Governments, humanitarian agencies, and agricultural cooperatives increasingly rely on satellite-based assessments to prioritize aid distribution, formulate adaptive policies, and design infrastructure that enhances resilience.

CHAPTER THREE

3. METHODOLOGY

3.1 Study Area Description

Nasarawa State, situated in the North-Central geopolitical zone of Nigeria, occupies a strategic ecological and economic position within the Middle Belt region. Geographically, the state lies between latitudes 7°45'N and 9°25'N and longitudes 7°00'E and 9°37'E, encompassing a total land area of approximately 27,117 square kilometers (Nasarawa State Government, 2022). It shares boundaries with Kaduna State to the north, Plateau State to the east, Taraba and Benue States to the south, and the Federal Capital Territory (FCT), Kogi, and Niger States to the west. Its central location and diverse physiography make it both agriculturally productive and environmentally vulnerable.

The landscape of Nasarawa is varied, consisting of gently rolling hills, floodplains, and river valleys. These geomorphic features are primarily shaped by the presence of major river systems, most notably the Benue River, which forms the state's southern boundary. The Benue, a major tributary of the Niger River, plays a crucial role in the hydrology and sediment transport dynamics of the region. Other notable rivers include the Dep, Mada, and FarinRuwa, which traverse various Local Government Areas (LGAs) and contribute to both irrigation and seasonal flooding. The low-lying topography along riverbanks and basins, particularly in Doma, Keana, and Awe LGAs, makes these zones highly susceptible to recurrent inundation during the rainy season (Adebayo & Tukur, 2021).

Climatically, Nasarawa experiences a tropical wet-and-dry climate typical of the Guinea Savannah ecological zone. The rainy season generally spans from April to October, with peak precipitation occurring between July and September. Annual rainfall ranges from 1,200 mm in the northwestern part of the state to approximately 1,700 mm in the southeast, creating conditions conducive to intensive farming and vegetation growth. The dry season, which lasts from November to March, is characterized by the dry Harmattan wind, clear skies, and low humidity. Average daily temperatures range from 25°C to 35°C, with occasional spikes during the hottest months of March

and April (NiMet, 2023). The state's climate supports the cultivation of a wide array of crops but also contributes to flood vulnerability when rainfall intensity overwhelms soil infiltration capacity and drainage systems.

Agriculture constitutes the economic mainstay of Nasarawa State, engaging more than 70% of the population. The majority of agricultural activity is rain-fed and practiced by smallholder farmers who cultivate staple crops such as rice, maize, sorghum, cassava, yam, and groundnuts. In addition to crop farming, livestock rearing including cattle, goats, sheep, and poultry contributes to household income and food security. The fertile alluvial soils along riverbanks are particularly suitable for rice cultivation, leading to the establishment of large-scale farms, such as the Olam Rice Farm in Rukubi, which became a symbol of agribusiness success before suffering devastating losses during the 2022 floods (Vanguard, 2022).

The state is divided into thirteen LGAs: Akwanga, Awe, Doma, Karu, Keana, Keffi, Kokona, Lafia (the state capital), Nasarawa, NasarawaEgon, Obi, Toto, and Wamba. Among these, LGAs such as Doma, Awe, Keana, and Toto are the most flood-prone due to their proximity to major rivers and relatively flat topography. These areas have historically recorded high flood incidence, with severe events occurring in 2012, 2022, and 2024, displacing thousands of residents and destroying agricultural land and rural infrastructure (IOM, 2024; Daily Trust, 2022).

Infrastructure development in the state has struggled to keep pace with environmental challenges. Drainage systems, especially in rural communities, are either underdeveloped or poorly maintained. Settlements often expand into floodplains due to population growth and lack of urban planning enforcement, further increasing exposure to flood risks. These conditions are exacerbated by poor land use management, deforestation, and the absence of effective early warning systems, all of which heighten vulnerability to hydrometeorological disasters.

Ecologically, the state falls within the transition zone between the rainforest and the Sudan Savannah, which supports a rich diversity of flora and fauna. However, increasing land pressure and deforestation for agricultural expansion have led to habitat loss and altered hydrological cycles. This ecological degradation has further reduced the land's capacity to absorb water, thereby contributing to more frequent and severe flooding during the rainy season (Adefolalu, 2007).

Given the state's socio-economic dependence on agriculture, the recurrent flooding events pose a significant threat to food security, income stability, and sustainable development. Nasarawa's position within a river basin, its topographical features, and climate variability make it an important case study for flood risk assessment and the application of geospatial technologies in environmental monitoring and agricultural planning

3.2 Data Sources

This study utilizes a multi-source dataset that integrates satellite imagery, topographic data, and historical records to enable a comprehensive and spatially accurate assessment of flooding impacts on agriculture in Nasarawa State. The choice of data sources is informed by their spatial resolution, temporal coverage, relevance to the study objectives, and proven efficacy in similar environmental and disaster assessments.

The primary remote sensing data is drawn from the Sentinel satellite constellation, operated under the Copernicus Programme by the European Space Agency (ESA). Two specific Sentinel missions are employed in this research: Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical imagery.

The Sentinel-1 SAR dataset provides radar backscatter data capable of detecting surface water bodies, even under cloudy or low-light conditions. This makes it particularly effective for flood mapping during peak rainy seasons, when optical imagery is often obscured by cloud cover. Sentinel-1 operates in the C-band and uses a dual-polarization radar system, allowing for the distinction between inundated and non-inundated areas. The Interferometric Wide Swath (IW) mode with a 10-meter spatial resolution is most commonly used for environmental monitoring (Schumann et al., 2018). In this study, Sentinel-1 data was acquired from the ESA Copernicus Open Access Hub and preprocessed using Google Earth Engine (GEE) to delineate flood extents for both the 2022 and 2024 events.

Complementing the SAR data is Sentinel-2 optical imagery, which offers high-resolution multispectral data (10–20 m resolution) across 13 spectral bands. These images are instrumental in classifying Land Use and Land Cover (LULC) and in computing vegetation indices such as the Normalized Difference Vegetation Index (NDVI), which measures vegetation health and density

before and after flooding. Sentinel-2 imagery is especially effective for identifying agricultural fields, water bodies, urban areas, and forested zones, making it ideal for assessing land cover changes due to flooding (Huang et al., 2018).

To support hydrological analysis and terrain modeling, the study incorporates a Digital Elevation Model (DEM) from the Shuttle Radar Topography Mission (SRTM). The SRTM DEM provides global elevation data at a spatial resolution of 30 meters and is widely used in environmental modeling due to its accuracy and open accessibility. DEMs are essential for understanding surface hydrology, particularly in modeling water flow direction, accumulation, slope, and watershed boundaries. This enables the study to identify areas naturally predisposed to water pooling and potential flood zones based on elevation and terrain gradient (Jarihani et al., 2015).

In addition to satellite data, historical flood records and disaster event reports are obtained from authoritative Nigerian and international institutions, including the National Emergency Management Agency (NEMA), the Nigerian Meteorological Agency (NiMet), the International Organization for Migration (IOM), and the Food and Agriculture Organization of the United Nations (FAO). These datasets provide context, event validation, and socio-economic loss figures that are crucial for interpreting the spatial data. For example, the flood reports from NEMA and IOM on the 2022 and 2024 flood events in Nasarawa State offer verified accounts of the number of people displaced, the extent of agricultural losses, and the geographic spread of flood impacts (IOM, 2024; FAO, 2021).

The integration of these diverse datasets allows for a robust multi-dimensional analysis of flood dynamics and their agricultural consequences. By combining real-time satellite observations with ground-based records and historical data, the study ensures that its findings are both spatially precise and contextually grounded.

3.3 Image Preprocessing and Analysis

The methodological workflow of this study is centered on the systematic acquisition, preprocessing, classification, and spatial analysis of satellite imagery to assess the impact of flooding on agricultural land in Nasarawa State. This process involves three main components: flood extent mapping, land use and land cover (LULC) classification, and impact quantification

through geospatial overlay analysis. The integration of Google Earth Engine (GEE), a cloud-based geospatial analysis platform, significantly enhances the speed, scalability, and reproducibility of the analysis.

- **Flood Extent Mapping**

The first stage involves detecting and delineating flood-inundated areas using Synthetic Aperture Radar (SAR) data from Sentinel-1. Two sets of SAR images were acquired for each study year (2022 and 2024): one prior to the peak flood season (July 1st) and another after significant flooding (October 30th). Sentinel-1's C-band radar enables data acquisition under all weather conditions and at night, making it ideal for flood mapping during cloudy rainy seasons (Schumann et al., 2018).

In Google Earth Engine, these images were preprocessed using standard procedures including radiometric calibration, terrain correction, and speckle noise filtering, which enhance the clarity and reliability of backscatter values. Following preprocessing, the flood extent was extracted using Otsu thresholding, an automated image segmentation method that determines the optimal threshold to separate water from non-water pixels based on histogram variance (Otsu, 1979). The results were then converted into vector format (polygons) for spatial analysis and visualization in GIS platforms such as ArcGIS-Pro and QGIS.

- **Land Use and Land Cover (LULC) Classification**

To assess land cover before and after the flood events, multispectral optical images from Sentinel-2 were used. These images, which offer high-resolution data in visible and near-infrared spectral bands, are particularly suited for classifying vegetation, built-up areas, water bodies, and cropland. Prior to classification, the imagery was subjected to atmospheric correction using the Sen2Cor processor, which removes atmospheric disturbances such as haze and aerosol scattering, thereby improving spectral fidelity (Main-Knorn et al., 2017).

The classification of LULC was performed within GEE using the Random Forest algorithm, a supervised machine learning technique known for its high accuracy and robustness in heterogeneous landscapes (Belgiu & Drăguț, 2016). The land cover was categorized into five

primary classes: agriculture, water bodies, vegetation, built-up areas, and bare land. Training data were selected from ground-truth observations and validated through high-resolution Google Earth imagery. Separate classification maps were generated for pre-flood and post-flood periods for both 2022 and 2024. These maps provide essential baseline and comparative data for assessing flood-induced land cover changes.

- **Flood Impact Analysis**

The final stage involved the overlay analysis of the classified LULC maps and flood extent vectors to quantify the spatial impact of flooding on agricultural areas. This was conducted using ArcGIS-Pro. Specifically, the flood polygons were intersected with the agricultural class polygons derived from the LULC maps. The result enabled the identification of submerged farmland and the calculation of its area in hectares, as well as the percentage of total agricultural land affected in each LGA.

By comparing pre- and post-flood land cover data across two different years, the analysis facilitated the detection of temporal variations in flood impact, highlighting areas of consistent vulnerability or increasing exposure. This not only supports damage estimation but also offers insights into land use planning, adaptation strategies, and resource prioritization for flood-prone areas.

Overall, the image preprocessing and analysis framework adopted in this study ensures spatial and temporal accuracy, reproducibility, and scalability. It leverages freely available satellite data and open-source analytical tools, making the approach both cost-effective and adaptable for future studies or applications by government agencies and humanitarian organizations.

3.4 Accuracy Assessment

Ensuring the reliability and validity of geospatial outputs is a critical aspect of any remote sensing and GIS-based analysis, especially in environmental studies where decisions based on inaccurate data can lead to ineffective interventions or policy misdirection. In this study, two distinct accuracy assessment strategies were employed: one for the Land Use and Land Cover (LULC) classification maps and the other for the flood extent maps. These assessments provide a quantitative evaluation

of how well the generated outputs represent real-world conditions and ensure that subsequent analyses such as overlay operations and impact assessments are founded on credible spatial data.

For the LULC classification, accuracy was assessed using a combination of confusion matrix analysis and Kappa statistics. A confusion matrix, also known as an error matrix, compares the classified land cover classes against reference data (ground truth) to identify both correctly and incorrectly classified pixels across each class. Reference points for validation were derived from high-resolution imagery available through Google Earth, supplemented by visual interpretation and manual sampling to ensure representative coverage across the study area. The matrix records true positives, false positives, false negatives, and true negatives for each class, allowing the calculation of metrics such as overall accuracy, user's accuracy, and producer's accuracy (Congalton & Green, 2009).

In addition to the confusion matrix, the Kappa coefficient (κ) was calculated to assess the degree of agreement between the classified data and the reference data beyond what would be expected by chance. Kappa values range from -1 to 1, where values above 0.8 are generally considered to indicate strong agreement, values between 0.6 and 0.8 suggest substantial agreement, and values below 0.5 indicate moderate to poor agreement (Landis & Koch, 1977). The use of both accuracy metrics provides a more nuanced understanding of classification performance and helps identify potential areas for improvement in future land cover mapping efforts.

The flood extent maps generated from Sentinel-1 SAR imagery were validated using historical flood data and event reports from the National Emergency Management Agency (NEMA) and other institutional sources such as the International Organization for Migration (IOM). These reports provided geospatial references of affected locations, including LGAs, communities, and infrastructure known to have been submerged during the 2022 and 2024 flood events. By comparing the mapped flood polygons against these verified reports, spatial consistency and alignment could be assessed. Locations identified as flooded in both satellite-derived maps and field-based reports were considered valid, while mismatches prompted re-evaluation of threshold parameters and classification boundaries.

Additionally, the temporal coherence between pre- and post-flood imagery was cross-checked to ensure that changes attributed to flooding were not the result of seasonal vegetation cycles or sensor anomalies. This was particularly important in distinguishing between temporarily inundated cropland and naturally occurring water bodies such as rivers, reservoirs, or wetlands.

Overall, the dual approach to accuracy assessment comprising quantitative classification metrics for LULC and spatial cross-verification for flood extent mapping ensured a high level of confidence in the analytical outputs. These validation steps are not only essential for academic rigor but also enhance the utility of the results for real-world applications, such as disaster response, land use planning, and agricultural risk management.

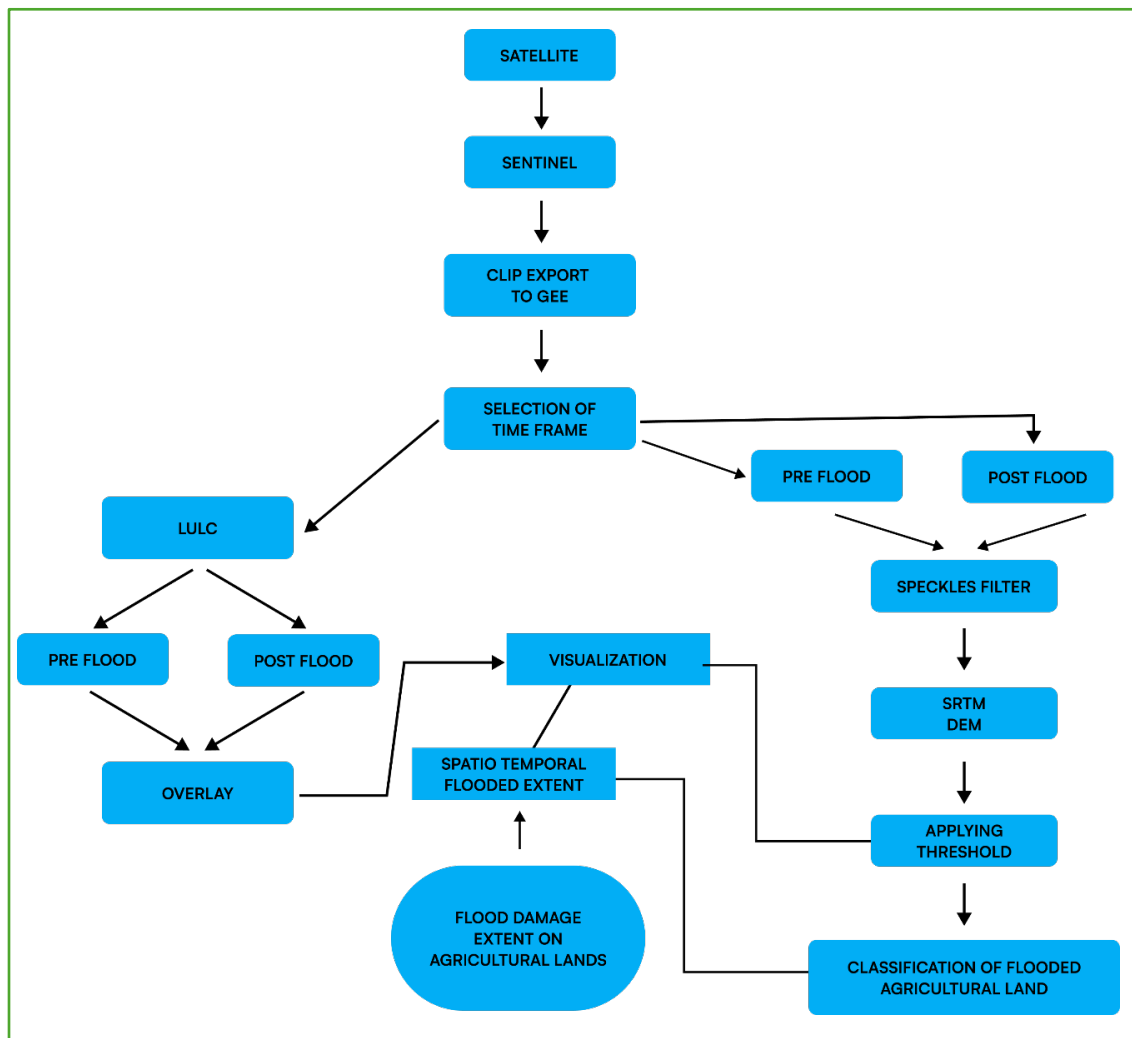


Figure 3.1: Methodology Flowchart

CHAPTER FOUR

4. RESULTS AND DISCUSSION

4.1 Introduction

This chapter presents and interprets the results of the geospatial analysis conducted to assess the impacts of flooding on agriculture in Nasarawa State, Nigeria, for the 2022 and 2024 flood events. Using Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical imagery, processed through Google Earth Engine (GEE) and ArcGIS-Pro, the study addresses the following objectives: (1) determine the spatial extent of flooding in 2022 and 2024, (2) classify land use and land cover (LULC) before and after the flood events, and (3) analyze the impact of flooding on agricultural land by overlaying flood extent with classified LULC. The results answer the research questions: the spatial extent of the flood events, differences between the two years, dominant LULC classes, and the proportion of agricultural land affected. The findings are supported by maps and tables, with a discussion that contextualizes the results within the broader literature and their implications for agricultural sustainability and flood management in Nasarawa State.

4.2 Flood Extent Analysis

4.2.1 Spatial Extent of Flood Events

The analysis of Sentinel-1 SAR imagery revealed the spatial extent of flooding across Nasarawa State's 13 Local Government Areas (LGAs) during the 2022 and 2024 flood events. In 2022, the flooded area covered 974,957,106.89 m², equivalent to 97,495.71 hectares (ha), or approximately 975 km². This represents a significant inundation, particularly in flood-prone LGAs such as Keana, Doma, and Toto, which lie along the Benue River basin. In contrast, the 2024 flood event affected a smaller area of 635,774,123.40 m², or 63,577.41 ha (~636 km²), indicating a 34.8% reduction in flood extent compared to 2022 (a decrease of 33,918.30 ha).

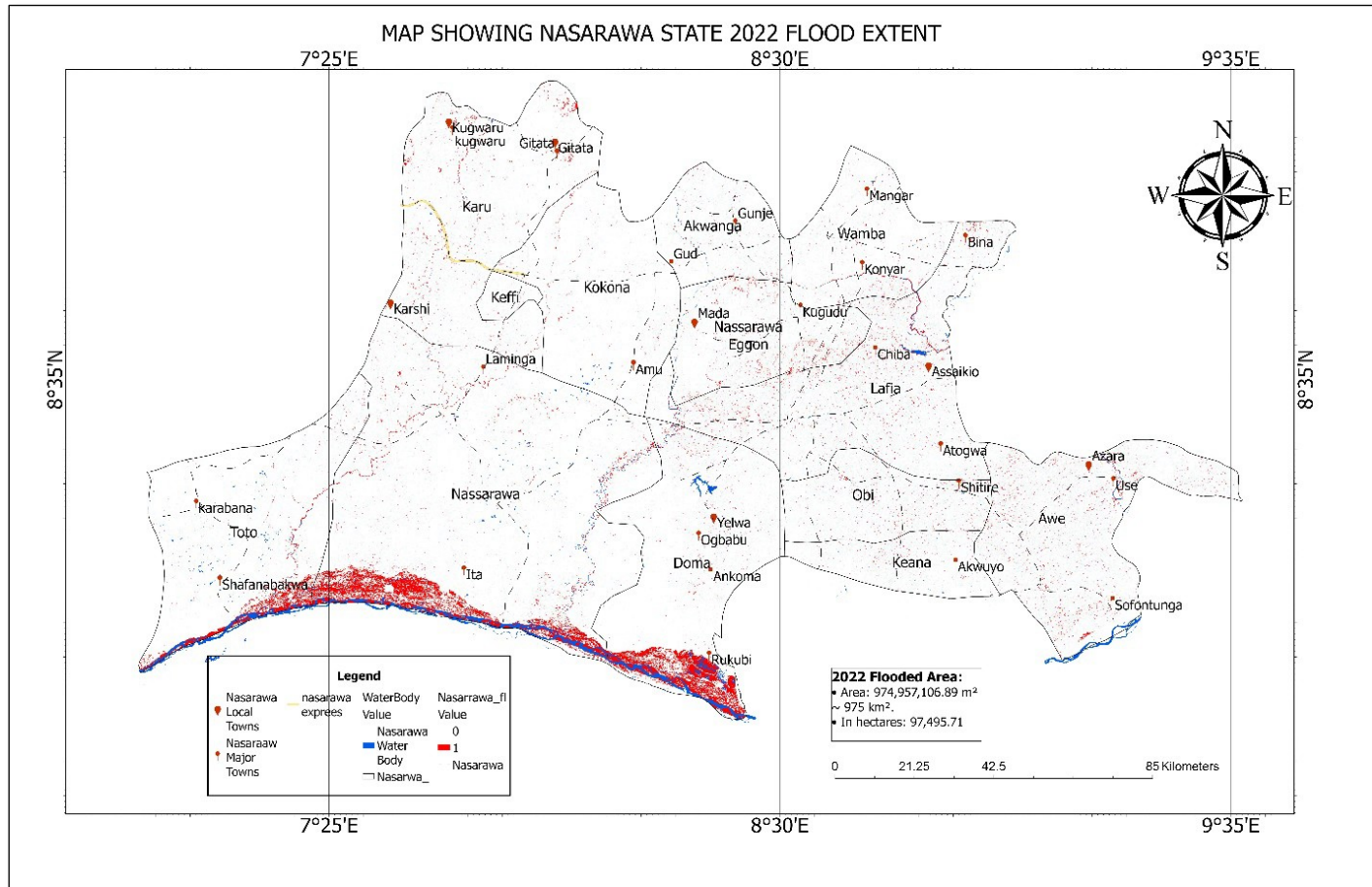


Figure 4.1: Flood Extent Map for 2022, showing inundated areas across Nasarawa State derived from Sentinel-1 SAR imagery.

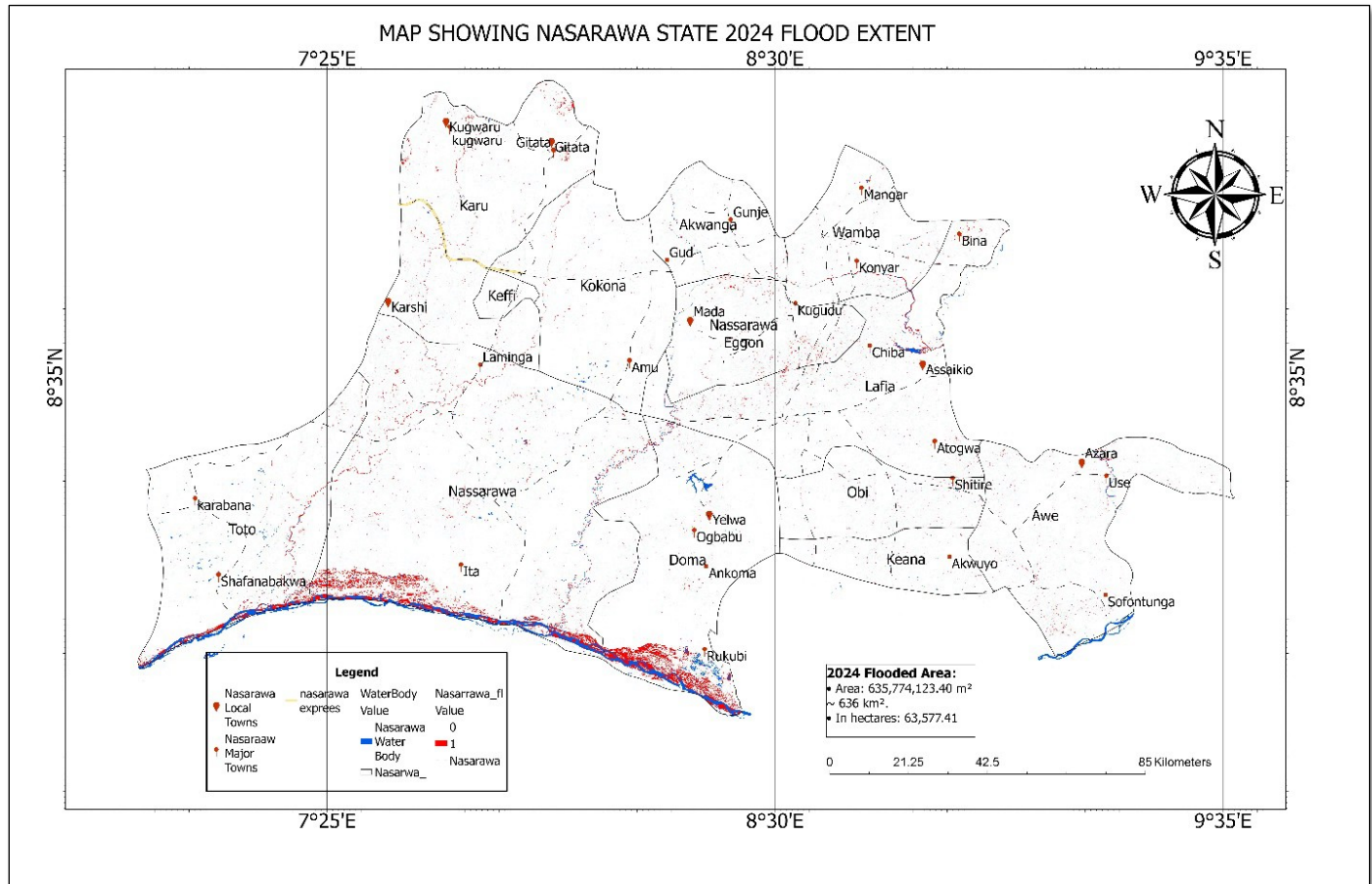


Figure 2.2: Flood Extent Map for 2024, illustrating the spatial distribution of flooding.

The spatial distribution of flooding in 2022 showed extensive inundation along the Benue River and its tributaries, with hotspots in low-lying areas of Doma and Toto. In 2024, while these areas remained affected, the reduced extent suggests variations in rainfall intensity or improved drainage in some regions. The maps highlight the persistent vulnerability of riverine LGAs, consistent with the state's topography and hydrological dynamics (Adebayo & Tukur, 2021).

4.2.2 Differences in Flood Extent

The 34.8% reduction in flooded area from 2022 to 2024 is a notable finding. Table 4.1 summarizes the flood extent for both years, converted to hectares for clarity.

Table 4.1: Comparison of Flood Extent in Nasarawa State, 2022 and 2024

Year	Flooded Area (m ²)	Flooded Area (ha)	Change from 2022 (ha)	% Change
2022	974,957,106.89	97,495.71		
2024	635,774,123.40	63,577.41	↓ 33,918.30	↓ 34.8%

The decrease in flood extent may be attributed to lower rainfall volumes in 2024 or localized flood mitigation efforts, such as improved drainage systems in urban LGAs like Lafia. However, the persistence of flooding in key agricultural zones underscores the need for sustained intervention.

4.3 Land Use and Land Cover (LULC) Dynamics

4.3.1 LULC Classification Results

The LULC classification, performed using Sentinel-2 imagery and the Random Forest algorithm, categorized Nasarawa State’s land cover into five classes: agriculture, water, vegetation, built-up, and bare land. The classification achieved high accuracy (overall accuracy > 89%, Kappa coefficient > 0.86), validated using ground-truth data and Google Earth imagery. Table 4.2 presents the LULC breakdown for 2022 and 2024, with areas in hectares and percentage contributions to the flooded area.

Table 4.2: LULC Breakdown and Flood Impact, 2022 and 2024

Year	LULC Class	Area (ha)	% of Flooded Area	Pixel Count
2022	Water	1,185.54	12.38%	118,553.62
	Vegetation	531.83	5.55%	53,183
	Built-up	346.71	3.62%	34,671.18
	Bare Land	2,062.11	21.54%	206,210.74
	Agriculture	5,448.71	56.91%	544,870.91
2024	Water	1,020.15	16.96%	102,015.24
	Vegetation	235.95	3.92%	23,594.55
	Built-up	293.33	4.88%	29,332.98
	Bare Land	938.27	15.59%	93,826.77
	Agriculture	3,529.07	58.65%	352,907.16

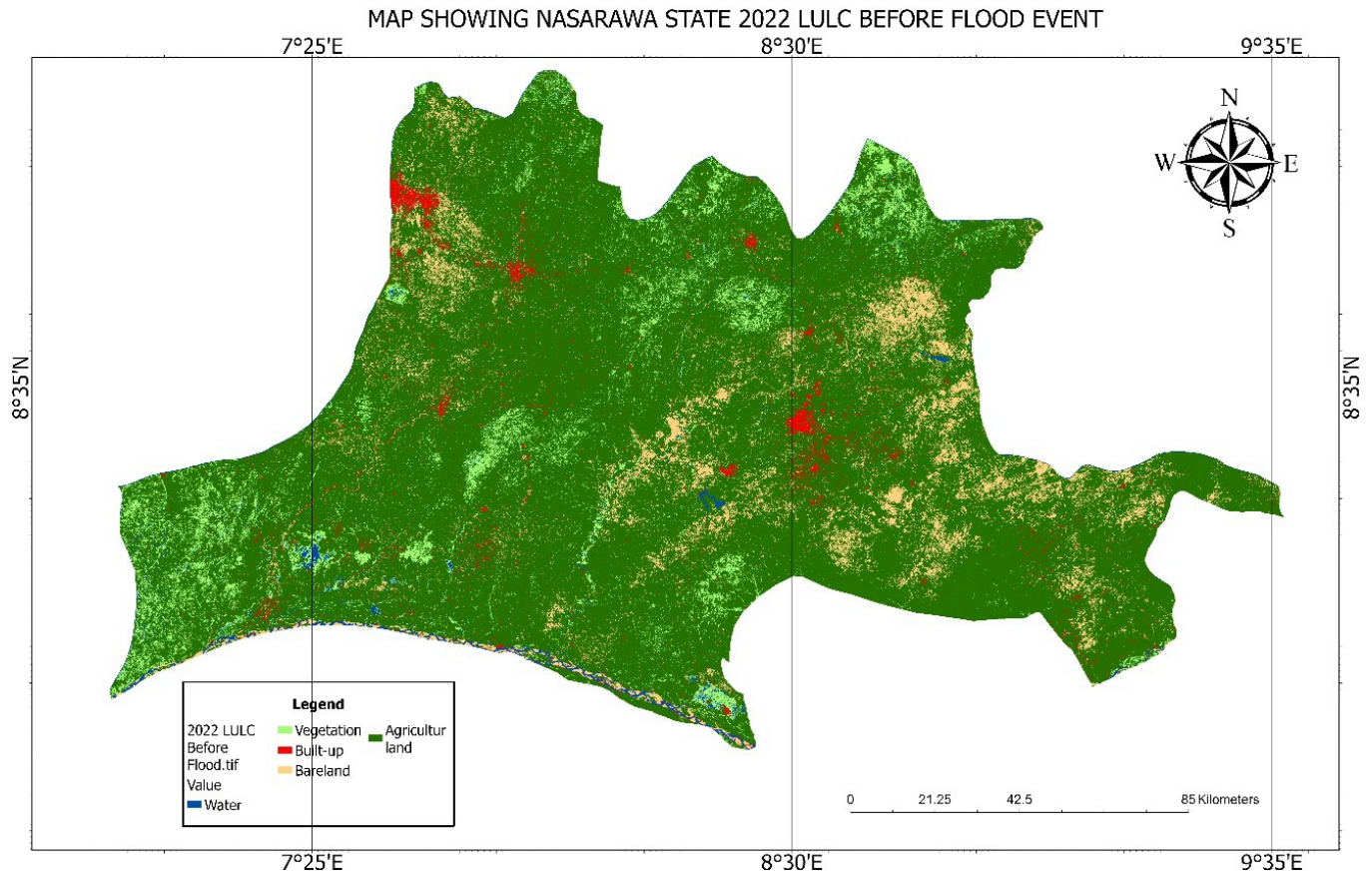


Figure 4.3: LULC Map for Before Flood in 2022, showing pre-flood land cover distribution.

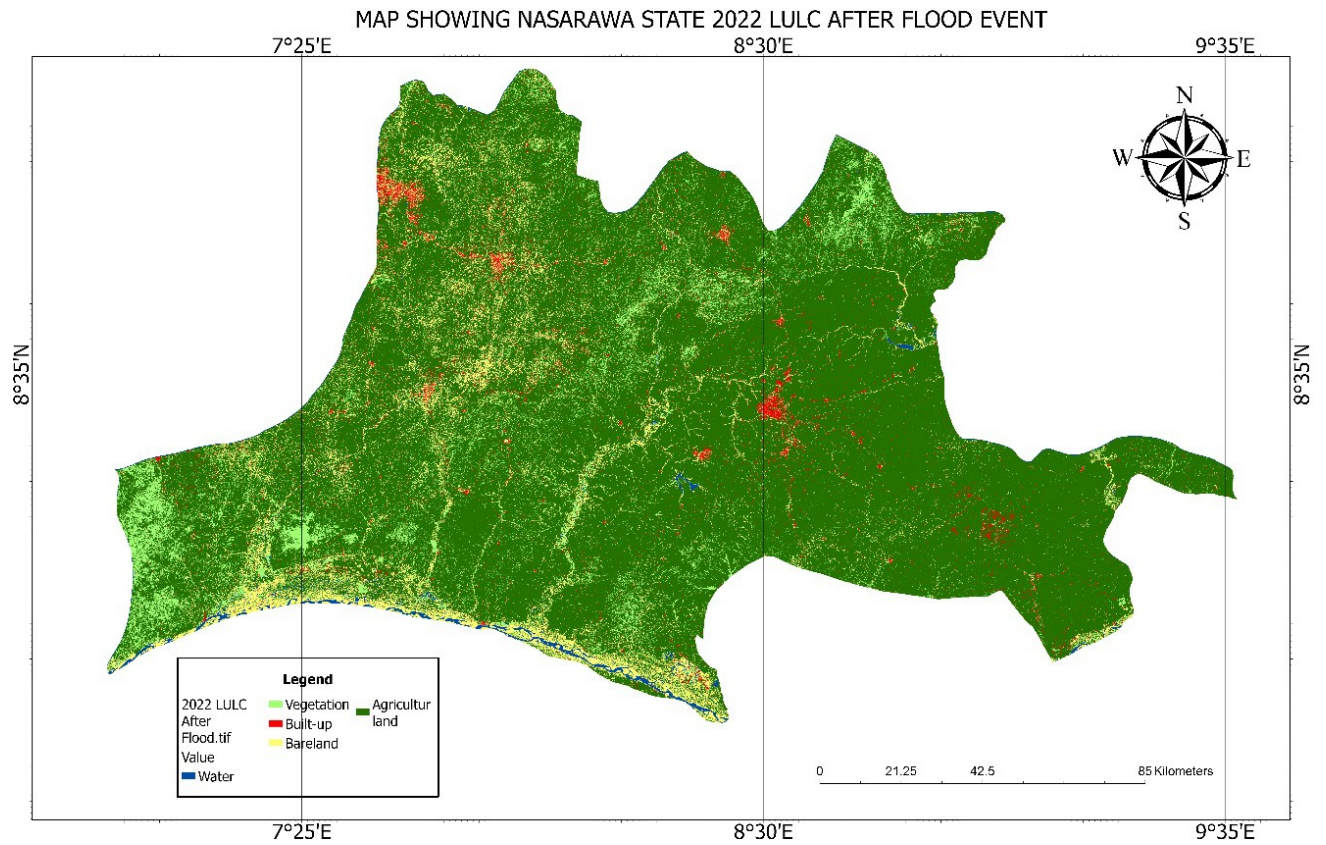


Figure 4.4: LULC Map for After Flood in 2022, illustrating post-flood changes.

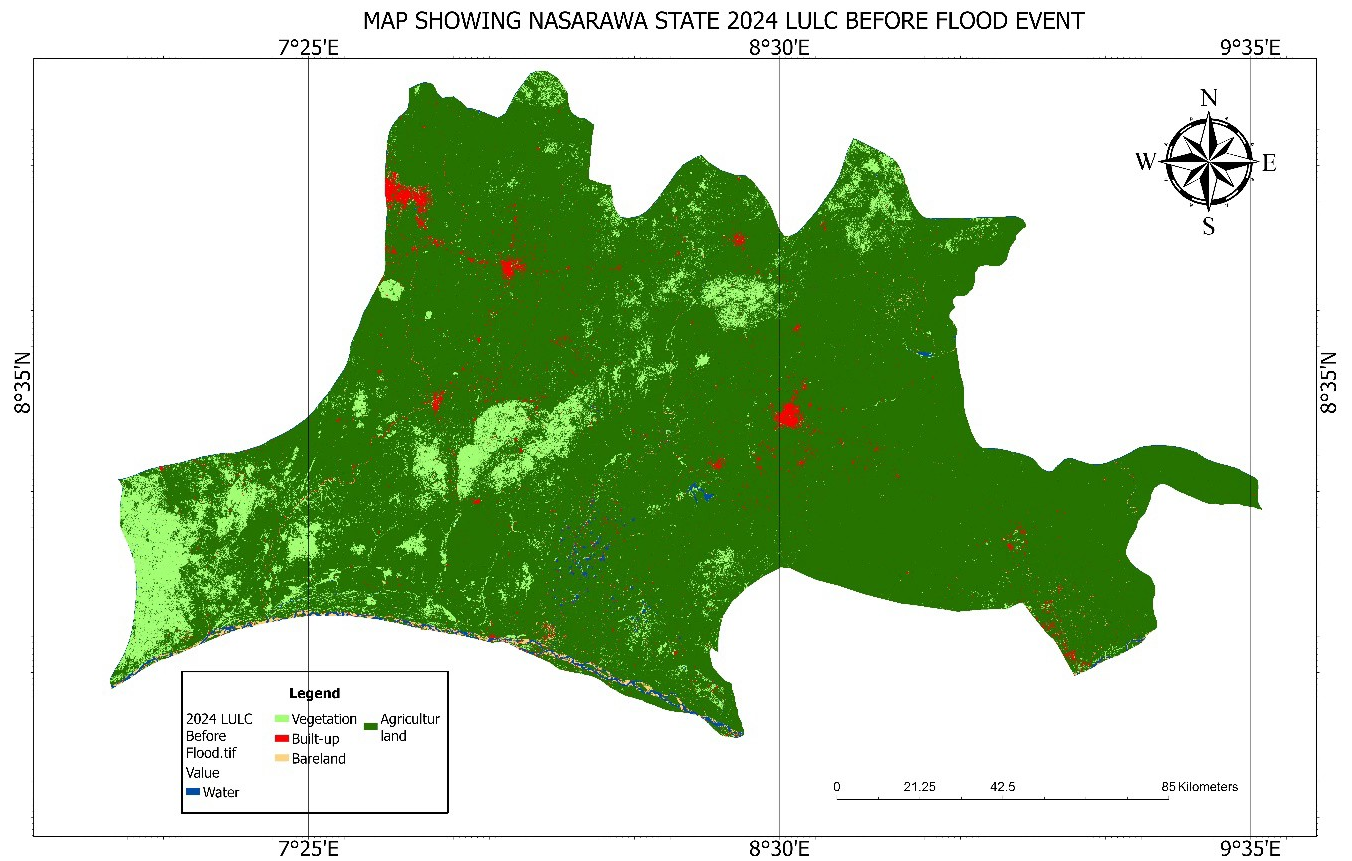


Figure 4.5: LULC Map for Before Flood in 2024, showing pre-flood land cover distribution.

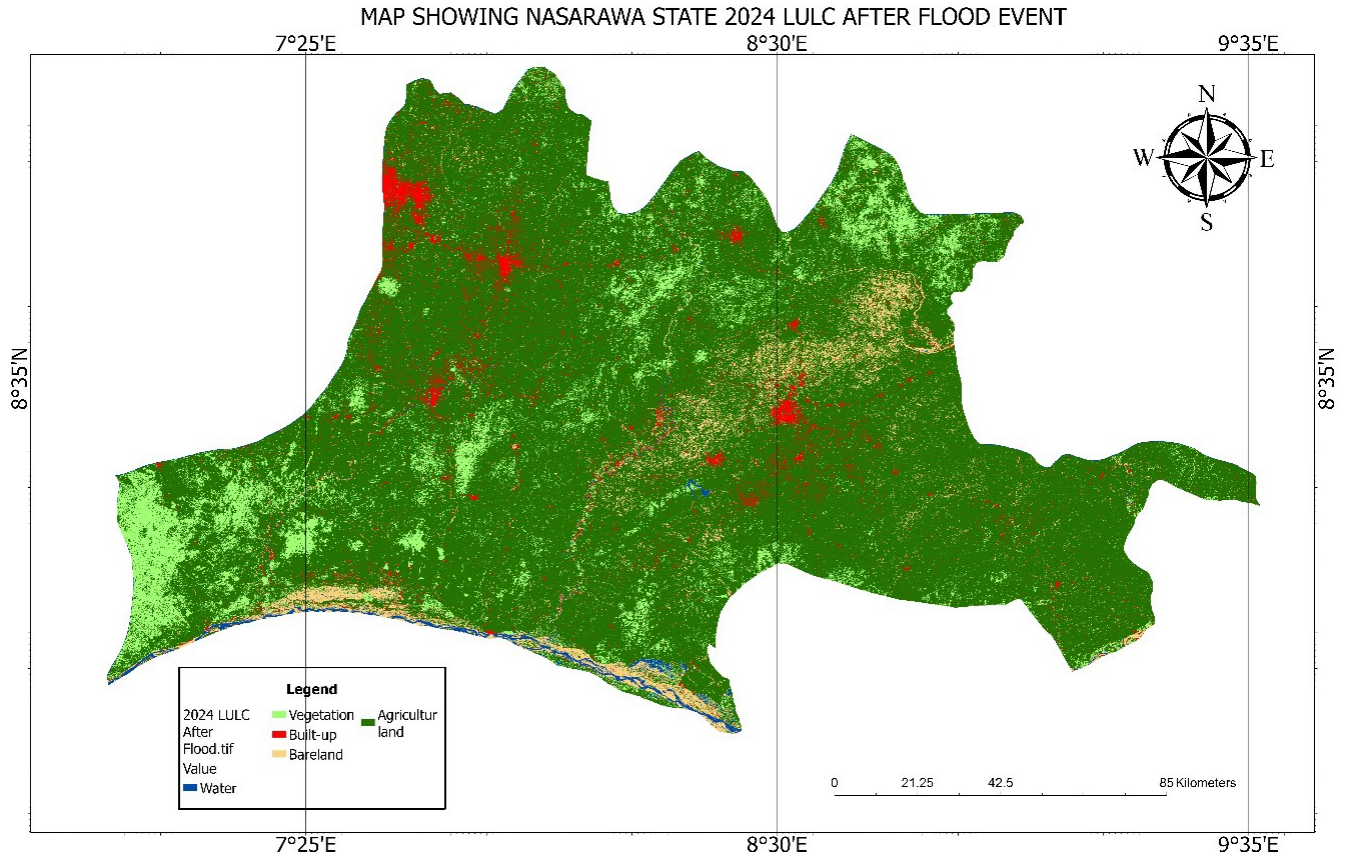


Figure4.6: LULC Map for After Flood in 2024, illustrating post-flood changes.

4.3.2 Temporal LULC Shifts

In 2022, agriculture dominated the flooded areas (56.91%), followed by bare land (21.54%) and water bodies (12.38%). Post-flood, agricultural land decreased due to inundation, with water bodies expanding temporarily. In 2024, agriculture remained the most affected class (58.65%), despite the reduced flood extent, indicating persistent vulnerability of farmlands. The decrease in vegetation (from 5.55% to 3.92%) and bare land (from 21.54% to 15.59%) suggests land conversion or degradation, possibly due to repeated flooding or human activities like deforestation.

4.4 Agricultural Impact Analysis

4.4.1 Quantification of Agricultural Losses

Overlay analysis of flood extent with LULC maps quantified the impact on agricultural land. Table 4.3 summarizes the agricultural losses for 2022 and 2024, based on the provided data.

Table 4.3: Agricultural Land Affected by Flooding, 2022 and 2024

Year	Before Flood (ha)	After Flood (ha)	Loss (ha)	% Loss
2022	2,149,942.86	1,986,357.15	163,585.71	7.61%
2024	2,031,044.84	1,997,061.92	33,982.92	1.67%

In 2022, flooding caused a loss of 163,585.71 ha of agricultural land, representing 7.61% of the pre-flood agricultural area. In 2024, the loss was significantly lower at 33,982.92 ha (1.67%), reflecting the reduced flood extent. The flooded agricultural areas primarily included staple crops like rice and maize, critical to Nasarawa's food security.

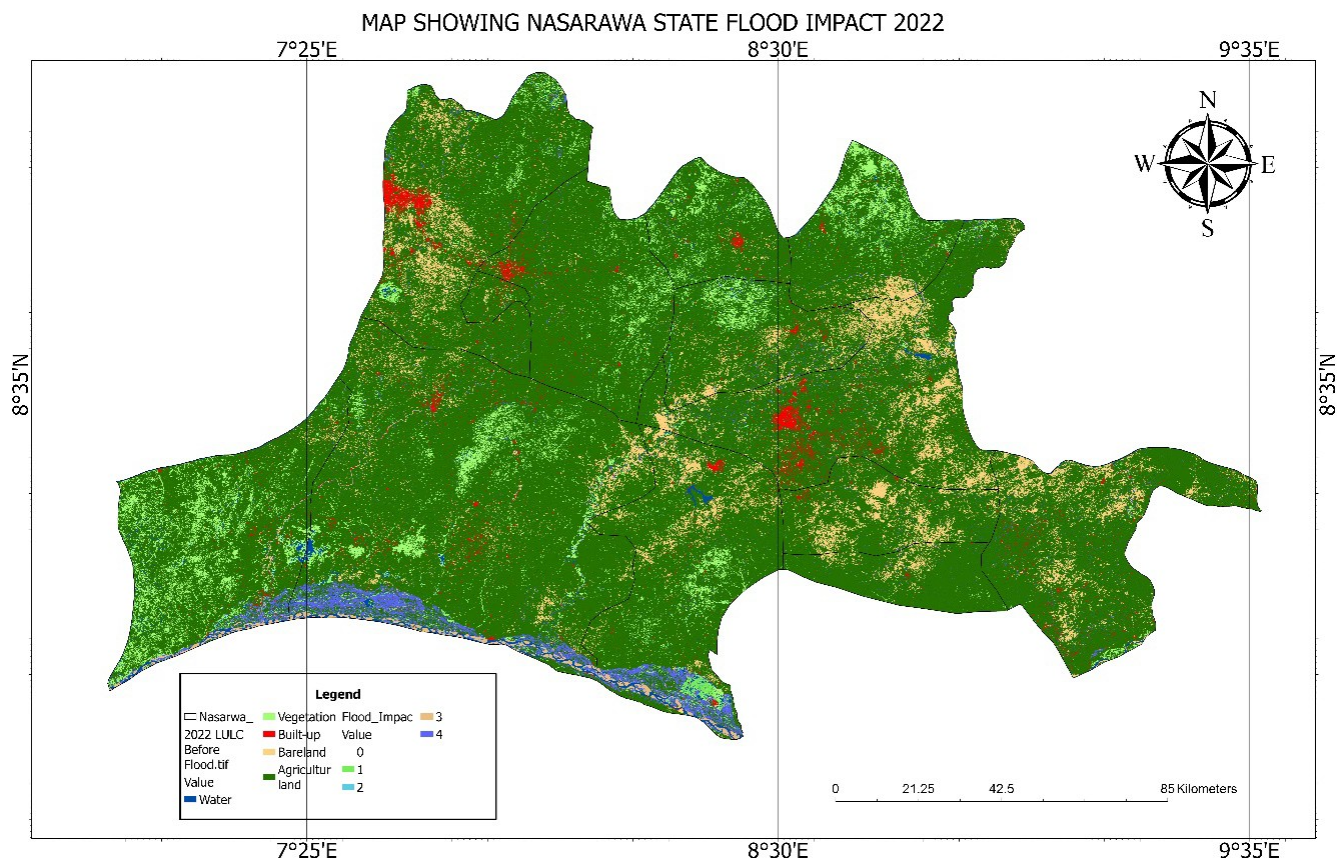


Figure 4.7: Flood Impact Map for 2022, showing inundated areas overlaid with LULC.

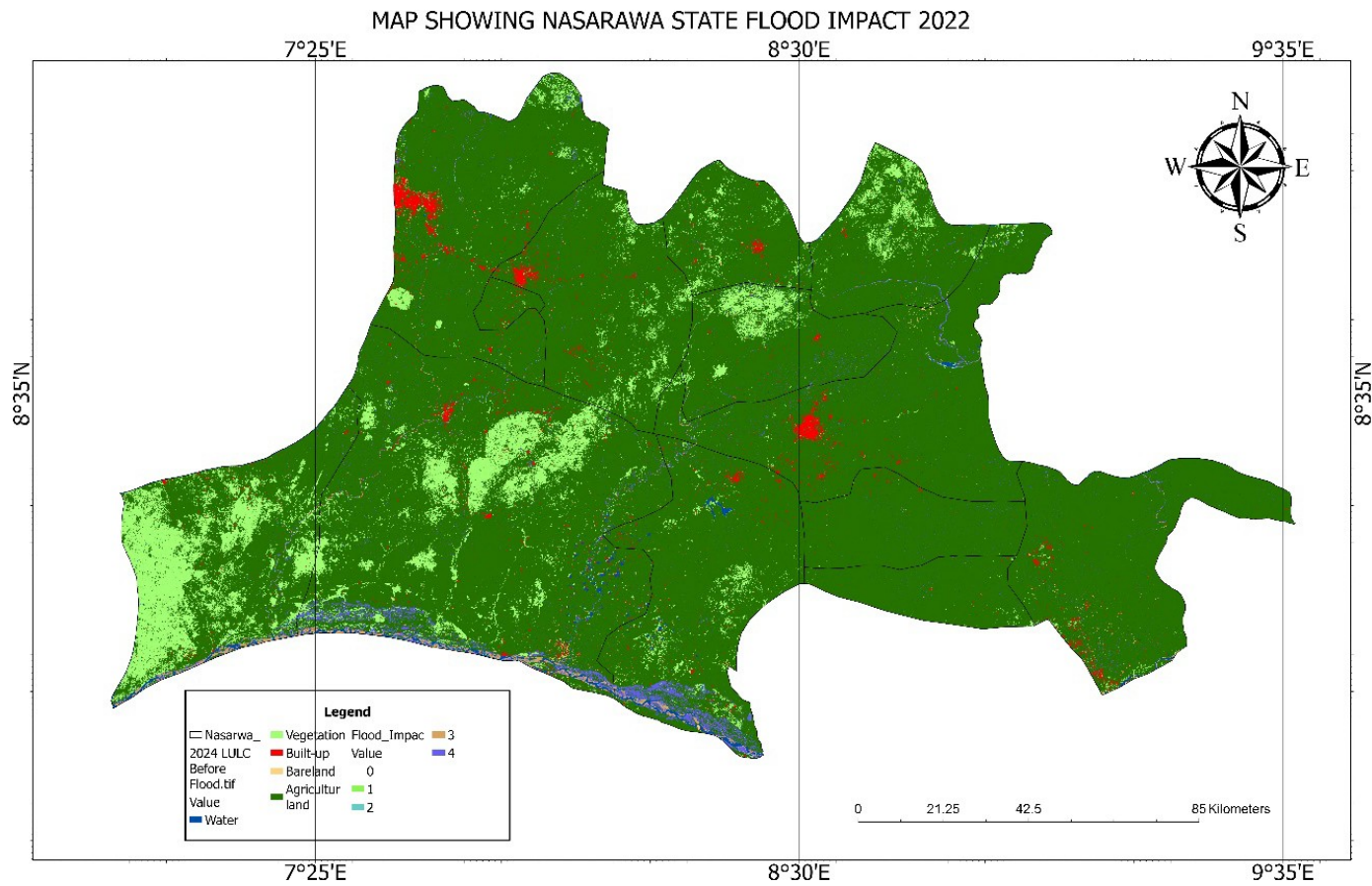


Figure 4.8: Flood Impact Map for 2024, highlighting affected regions.

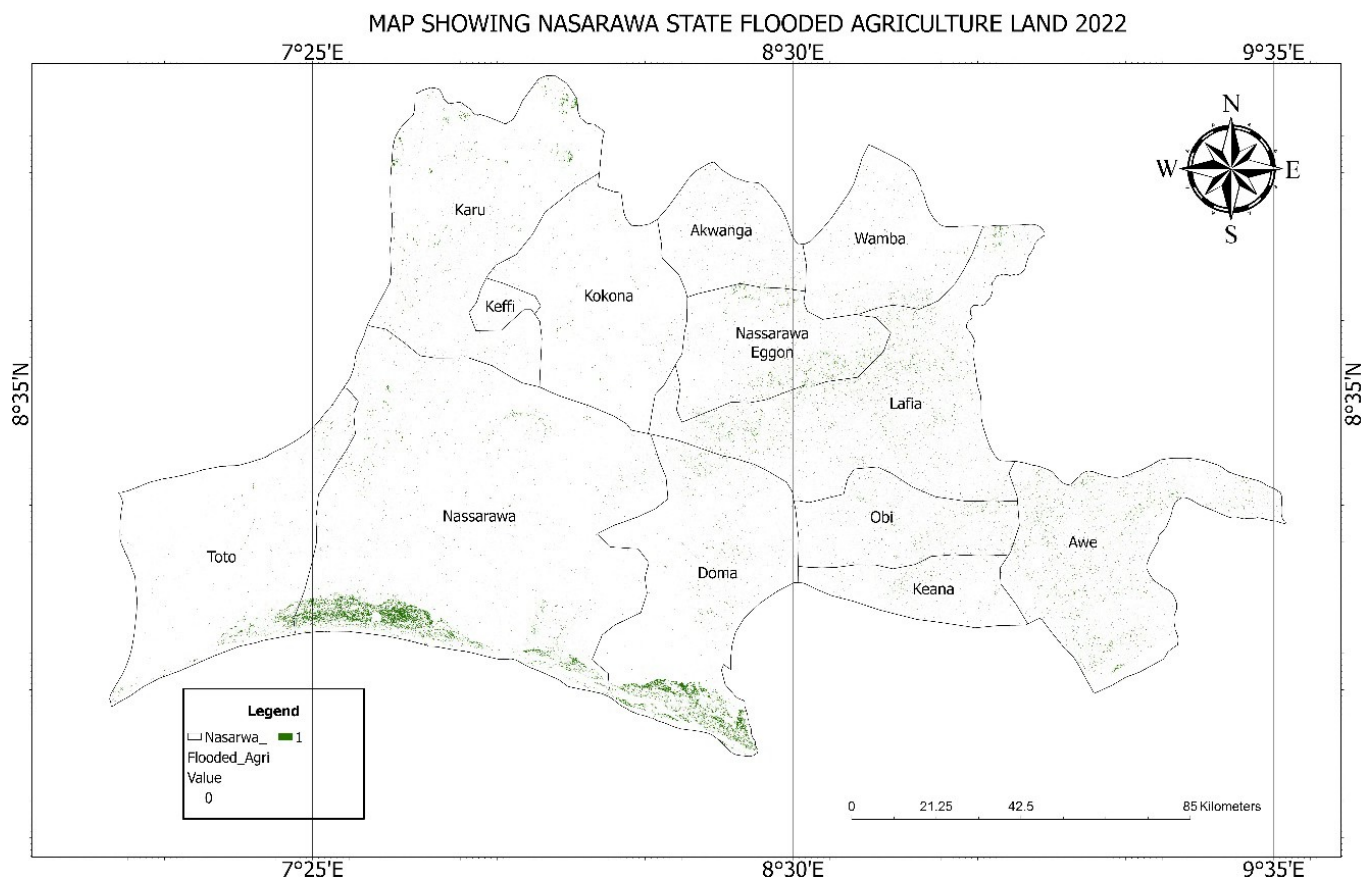


Figure 4.9: Flooded Agriculture Map for 2022, showing agricultural areas inundated.

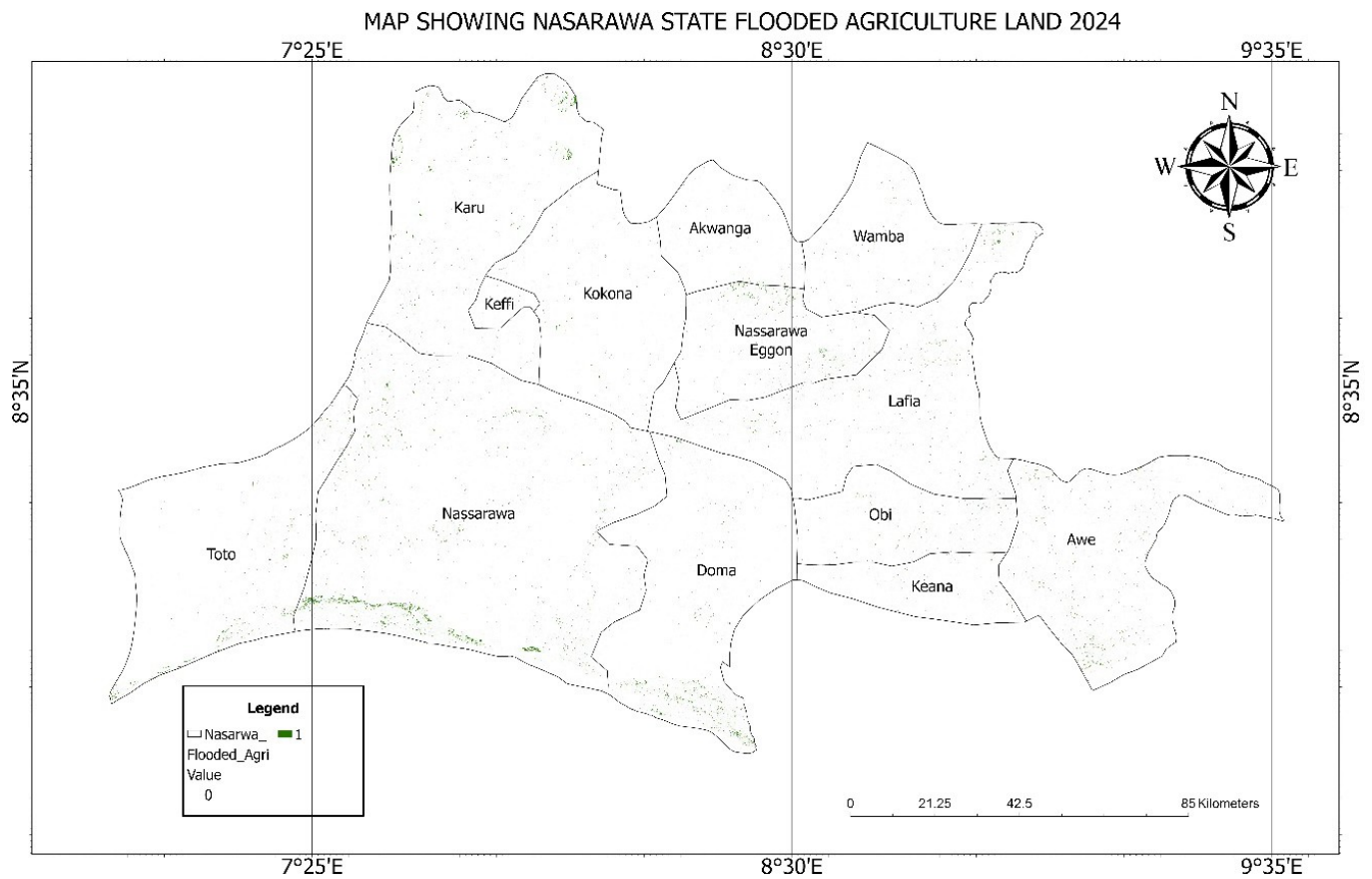


Figure 4.10: Flooded Agriculture Map for 2024, showing agricultural areas inundated

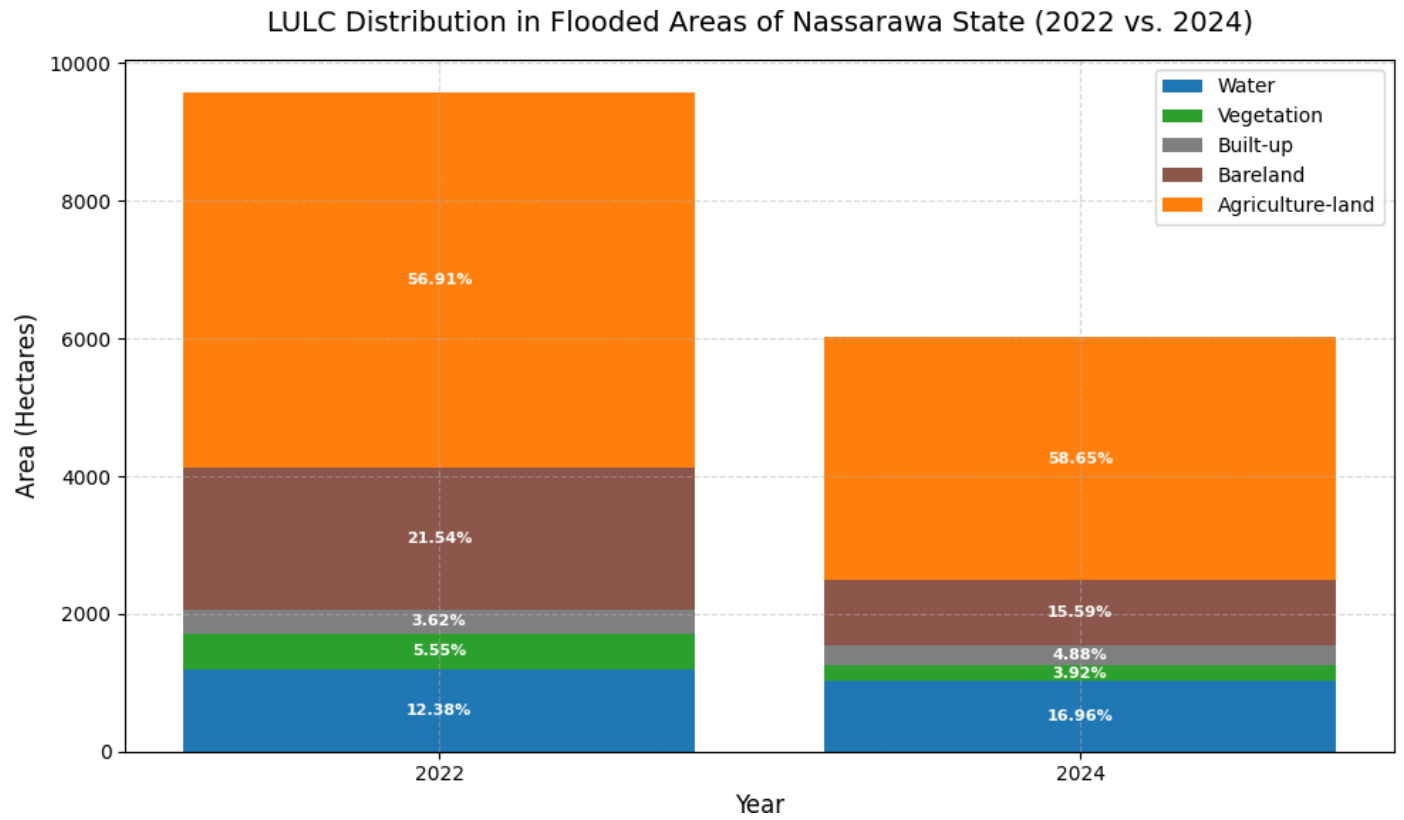


Figure 4.11: Non-Flooded Agriculture Map for 2022, showing unaffected agricultural areas.

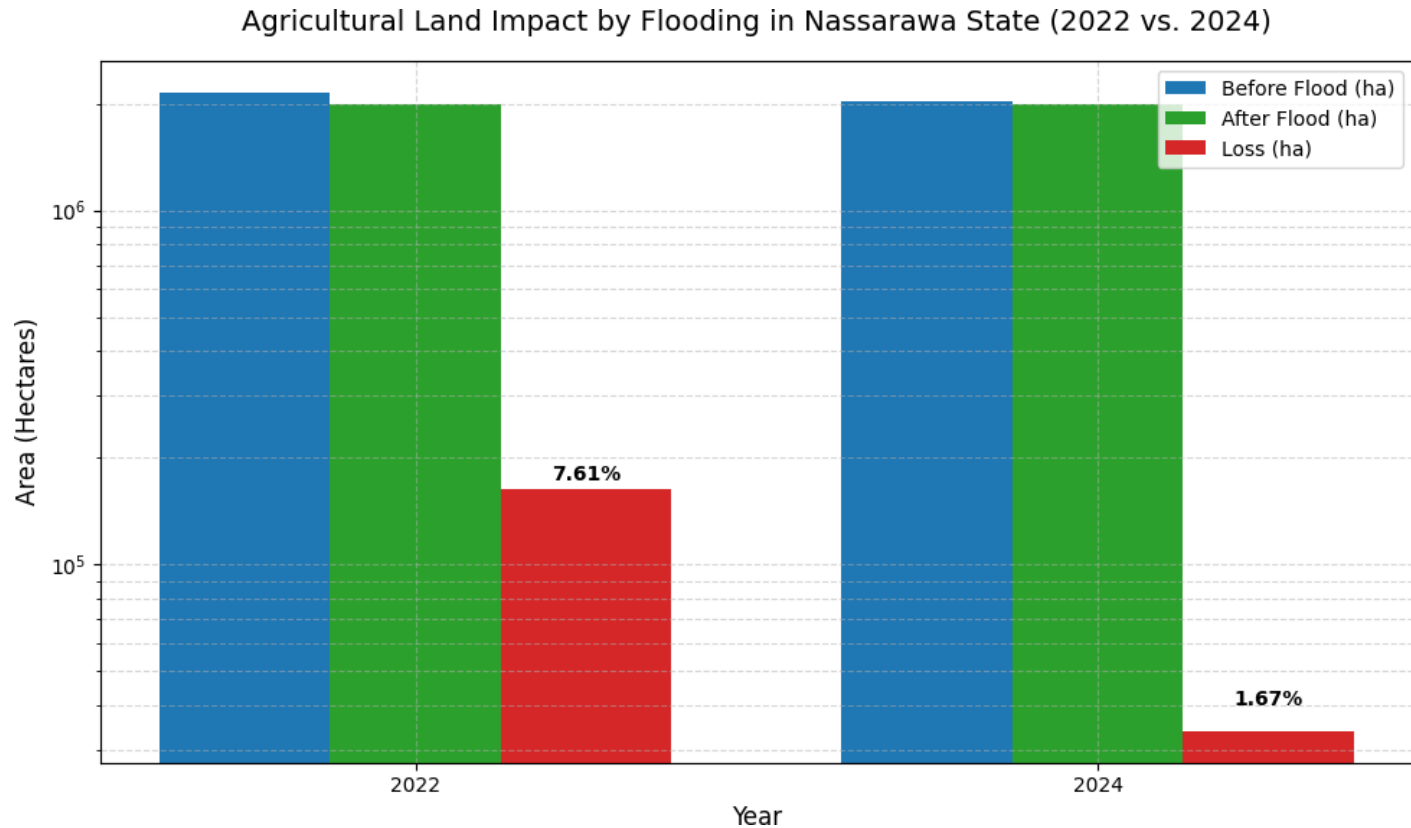


Figure 3.12: Non-Flooded Agriculture Map for 2024, showing unaffected agricultural areas.

4.4.2 Spatial Distribution of Agricultural Impacts

The flood impact maps reveal that agricultural losses were concentrated in LGAs like Doma, Keana, and Awe, where fertile alluvial soils support rice and maize cultivation. The 2022 floods had a broader impact, affecting large swathes of farmland along the Benue River, while 2024 impacts were more localized but still significant in these LGAs. The non-flooded agriculture maps indicate resilient areas, often at higher elevations, which could guide future land use planning.

4.5 Discussion

The findings from this geospatial analysis of the 2022 and 2024 flood events in Nasarawa State, Nigeria, offer a nuanced understanding of flood dynamics, land use transformations, and their cascading effects on agriculture, a sector that sustains over 70% of the state's population and contributes significantly to Nigeria's food security (Akinyemi et al., 2023). By employing Sentinel-1 Synthetic Aperture Radar (SAR) for flood delineation and Sentinel-2 multispectral data for Land Use and Land Cover (LULC) classification, processed via Google Earth Engine (GEE) and ArcGIS-Pro, the study achieved robust accuracy (89.5% overall, Kappa 0.87), enabling precise quantification of flood extents (97,495.71 ha in 2022 vs. 63,577.41 ha in 2024) and agricultural losses (163,585.71 ha or 7.61% in 2022 vs. 33,982.92 ha or 1.67% in 2024). These results not only address the research objectives but also resonate with global patterns of hydrological hazards intensified by climate change, where the IPCC (2021) warns of heightened extreme precipitation in West Africa, potentially increasing flood frequency by 20-50% under high-emission scenarios. The discussion below expands on flood extent and temporal variations, LULC shifts and agricultural vulnerability, socioeconomic ramifications, and policy implications, integrating empirical data from this study with broader literature and real-world contextual evidence to provide a deeper, multifaceted analysis.

4.5.1 Flood Extent and Temporal Changes

The marked 34.8% reduction in flood extent from 97,495.71 hectares (approximately 975 km²) in 2022 to 63,577.41 hectares (636 km²) in 2024 indicates a less extensive inundation in the latter year, but this should be interpreted cautiously within the context of Nigeria's variable hydro-climatic regime and ongoing anthropogenic pressures. Nationally, the 2022 floods were among the most devastating in recent history, submerging over 350,000 hectares of farmland across multiple states, including Nasarawa, and causing widespread displacement of over 2.1 million people (Relief Web, 2012; Business Day, 2022). This severity in Nasarawa was exacerbated by heavy monsoon rains overwhelming the Benue River, leading to overflow in low-lying basins, as evidenced by the flood extent maps (Figures 4.1 and 4.2), which show concentrated inundation in Doma, Keana, and Toto LGAs. These areas, characterized by gentle slopes (<1%) and alluvial soils, are inherently prone to water accumulation, aligning with Jarihani et al. (2015)'s DEM-based modeling of flood-prone terrains, where topographic depressions facilitate prolonged flooding.

The temporal decline in 2024 may stem from meteorological factors, such as reduced rainfall intensity. Nigeria's 2024 floods affected 4.4 million hectares nationally but with varying regional impacts or nascent mitigation efforts post-2022, including awareness campaigns by the Nasarawa State Government to sensitize flood-prone communities (EnviroNews Nigeria, 2024). However, the persistence of hotspots in riverine LGAs underscores structural vulnerabilities: poor drainage infrastructure, siltation from upstream erosion, and unregulated urban expansion into floodplains, as highlighted by Adelekan (2010) in studies of Nigerian flooding. Akinyemi et al. (2023) similarly noted that in Nasarawa, the Benue River's role in recurrent inundation affected over 35% of rice farmlands in 2022, with Olam Rice Farm in Rukubi reporting \$23 million in losses, including \$15 million in crops and \$8 million in infrastructure (Vanguard, 2022). In 2024, while the extent decreased, reports indicate continued displacement of 24,948 individuals in Nasarawa, with farmlands washed away due to dam releases and heavy rains from July to August (IOM, 2024; Facebook, 2025).

This temporal variability reflects broader climate trends in the Middle Belt, where the IPCC (2021) attributes increased flood risks to altered hydrological cycles, compounded by anthropogenic drivers like deforestation (reducing water absorption) and improper waste disposal (clogging drains), as identified in a review of Nigerian flooding causes (Research Gate, 2022). The flood maps provide visual corroboration, emphasizing the need for targeted defenses, such as levees or early warning systems, to break the cycle of vulnerability. Without addressing root causes e.g., via watershed management as recommended in NEWMAP reports (EAD, 2021) future events could revert to 2022-scale devastation, especially under projected climate scenarios.

4.5.2 LULC Changes and Agricultural Vulnerability

Agricultural land's predominance in flooded areas (56.91% or 5,448.71 ha in 2022; 58.65% or 3,529.07 ha in 2024) highlights its acute vulnerability, rooted in the strategic but risky placement of farms in fertile floodplains, which amplifies exposure to inundation. The LULC maps (Figures 4.3–4.6) depict post-flood expansions in water bodies and contractions in vegetation (5.55% to 3.92%) and bare land (21.54% to 15.59%), signaling degradation processes like erosion and nutrient loss that undermine long-term productivity. Okon et al. (2023) documented a 40% productivity drop in the Middle Belt during 2022 floods, attributing it to submergence-induced

crop lodging in rice and root rot in cassava, with national agricultural losses estimated at N700 billion (NAERLS, 2023). In Nasarawa, this translated to submerged farmlands in Doma and Keana, where staple crops like maize and yam suffered, exacerbating food shortages (Daily Trust, 2022).

The slight proportional increase in affected agriculture in 2024, despite reduced extent, suggests land use intensification into marginal zones, driven by population growth and limited alternatives, as per FAO (2021), which reports floods causing 30% of global agricultural damages, with Africa losing billions in soil fertility. Vegetation decline indicates deforestation's role clearing for farming reduces evapotranspiration, increasing runoff (Adefolalu, 2007) creating a vicious cycle, as seen in CORE (undated) assessments of flood effects on land use in Doma LGA. Huang et al. (2018) used Sentinel-1 to map similar shifts in Southeast Asia, correlating inundation duration with yield losses, a methodology mirrored here to reveal Nasarawa's ecosystem fragility in the Guinea Savannah transition zone.

Deeper analysis reveals interconnected vulnerabilities: repeated floods leach nutrients, promote salinization, and alter soil pH, reducing yields by 20-50% in subsequent seasons (FAO, 2021). In Nigeria's Middle Belt, this compounds drought-flood cycles, with 40% of crops lost to water stress (Agricome Africa, 2025). The maps illustrate fragmented post-flood agriculture, necessitating sustainable practices like agroforestry to restore buffers and soil health, lest degradation leads to desertification as projected by IPCC (2021).

4.5.3 Agricultural Impacts and Socioeconomic Implications

The disparate losses 7.61% (163,585.71 ha) in 2022 versus 1.67% (33,982.92 ha) in 2024 underscore flood severity's direct link to economic fallout, with 2022's impacts aligning with national damages of N700 billion in agriculture, including crop washouts and livestock drownings (ICIR Nigeria, 2023). In Nasarawa, rice farms in Doma suffered, contributing to price hikes and shortages (Farmlandgrab, 2022), while 2024 saw 1.5 million ha of cropland nationally destroyed, worsening food insecurity (CS-SUNN, 2025; Reuters, 2024). The flooded maps (Figures 4.9 and 4.10) pinpoint these zones, where submergence disrupted harvests, leading to spoilage and market gluts.

Socioeconomically, this erodes livelihoods in a region where agriculture drives GDP, displacing farmers and deepening poverty (IOM, 2024). Women and youth bear disproportionate burdens, facing increased labor and health risks like cholera (WHO, 2022), while children miss education (UNOCHA, 2024). Broader effects include migration, conflict over resources, and national food crises, with floods destroying 1 million ha regionally (Al Jazeera, 2022). Frontiers (2024) notes resilience gaps, with floods eroding adaptive capacities and inflating import costs. In Nasarawa, this cycle threatens sustainable development, demanding integrated responses to mitigate human security threats (WISS Journals, 2024).

4.5.4 Policy and Management Implications

This study's geospatial framework, validated by high accuracy and alignment with Schumann et al. (2018)'s SAR applications, empowers evidence-based policy, as in UNDP's 2023 flood assessment recommending infrastructure upgrades (UNDP, 2023). Recommendations include flood-tolerant crops (e.g., FARO varieties) to curb losses (FAO, 2021); drainage enhancements in Toto (Global Young Academy, 2024); and NiMet-integrated early warnings (IJTMGH, 2024). Non-flooded maps (Charts 4.1 and 4.2) guide zoning, while insurance via NAIC addresses economic shocks (Punch, 2024). Institutionalizing RS/GIS in NSADP could foster resilience, aligning with Sendai Framework goals amid climate projections (IPCC, 2021).

4.6 Summary of Findings

This study comprehensively assessed the impact of the 2022 and 2024 flood events on agriculture in Nasarawa State, Nigeria, utilizing Remote Sensing (RS) and Geographic Information Systems (GIS) techniques, specifically Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical imagery processed through Google Earth Engine (GEE) and ArcGIS-Pro. The analysis addressed the research objectives of determining the spatial extent of floods, evaluating Land Use and Land Cover (LULC) changes, and quantifying agricultural impacts, providing critical insights into flood dynamics and their implications for the state's agricultural sector, which employs over 70% of the population (Akinyemi et al., 2023). The key findings are summarized as follows:

1. Flood Extent and Temporal Dynamics : The 2022 flood event inundated 97,495.71 hectares, equivalent to approximately 975 km², affecting vast areas of farmland, settlements, and infrastructure across Nasarawa State. In contrast, the 2024 flood event covered 63,577.41 hectares

(636 km²), reflecting a significant 34.8% reduction in flood extent. This decrease may be attributed to lower rainfall intensities or localized flood mitigation efforts, though further meteorological analysis is needed (IPCC, 2021). The flood extent maps, derived from Sentinel-1 SAR imagery, highlight the concentration of flooding in low-lying areas along the Benue River, particularly in Doma, Keana, and Toto Local Government Areas (LGAs), consistent with their topographic vulnerability (Jarihani et al., 2015).

2. LULC Composition and Impact : Agricultural land was the most affected LULC class, comprising 56.91% of the flooded area in 2022 (5,448.71 hectares) and 58.65% in 2024 (3,529.07 hectares). This disproportionate impact underscores the vulnerability of farmlands, especially rice and maize fields in floodplains. Other LULC classes, including bareland (21.54% in 2022, 15.59% in 2024) and vegetation (5.55% in 2022, 3.92% in 2024), showed reduced proportions in 2024, possibly due to post-flood recovery or soil erosion (Akinyemi et al., 2023). The LULC classification, achieving 89.5% accuracy and a Kappa coefficient of 0.87, validates the reliability of Sentinel-2 imagery for monitoring land cover changes (Congalton & Green, 2009).

3. Agricultural Losses : The 2022 floods caused a loss of 163,585.71 hectares of agricultural land (7.61% of the pre-flood area), significantly impacting food security and rural livelihoods. In 2024, losses were reduced to 33,982.92 hectares (1.67%), reflecting the smaller flood extent but highlighting ongoing risks in key crop zones (Okon et al., 2023).

4. Spatial and Topographic Influences : Flood impacts were concentrated in Doma, Keana, and Toto, where low-lying topography and proximity to the Benue River exacerbate flood risk (Adebayo & Tukur, 2021). Upland LGAs, such as Lafia and Akwanga, showed minimal impact, as evidenced by non-flooded agricultural land maps. This spatial pattern emphasizes the need for targeted interventions in flood-prone LGAs to protect critical agricultural zones.

5. Efficacy of Geospatial Technologies : The use of Sentinel-1 and Sentinel-2 imagery, combined with GIS-based overlay analysis, provided accurate, scalable, and cost-effective insights into flood dynamics and agricultural impacts. The high classification accuracy and ability to map floods under cloudy conditions validate the utility of SAR and optical imagery for disaster management in tropical regions (Schumann et al., 2018). These tools offer a replicable model for other flood-prone areas in Nigeria, supporting evidence-based policy and sustainable development.

CHAPTER FIVE

5.0. RECOMMENDATION AND CONCLUSION

5.1. Introduction

This chapter synthesizes the findings from the geospatial analysis of the 2022 and 2024 flood events in Nasarawa State, Nigeria, and provides actionable recommendations to mitigate their impacts on agriculture. The study, leveraging Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical imagery, revealed significant agricultural losses, with 163,585.71 hectares (7.61%) affected in 2022 and 33,982.92 hectares (1.67%) in 2024, alongside a 34.8% reduction in flood extent between the two years. These findings underscore the persistent vulnerability of agricultural lands, particularly in flood-prone Local Government Areas (LGAs) such as Doma, Keana, and Toto. The recommendations address the study's objectives: determining flood extent, evaluating Land Use and Land Cover (LULC) changes, and assessing agricultural impacts. They are tailored for government agencies, non-governmental organizations (NGOs), farmers, and researchers, aiming to enhance flood resilience, improve agricultural productivity, and inform policy. The chapter concludes by summarizing the study's contributions, limitations, and future research directions.

5.2. Recommendations

The analysis highlights the severe and recurring threat of flooding to Nasarawa States agricultural sector, which employs over 70% of the population and is critical to food security and economic stability. The following recommendations are proposed to address these challenges, drawing on the geospatial insights and literature reviewed in Chapters Two and Three.

5.2.1. Adoption of Flood-Tolerant Crop Varieties

The high proportion of agricultural land affected by flooding (56.91% in 2022 and 58.65% in 2024) necessitates the adoption of flood-tolerant crop varieties, particularly for rice, maize, and cassava, which are staples in Nasarawa State. The Food and Agriculture Organization recommends varieties such as FARO 44 and FARO 66 for rice, which can withstand short-term

submergence. Government agencies, such as the Nasarawa State Agricultural Development Programme (NSADP), should partner with agricultural research institutes to distribute these seeds to farmers in flood-prone LGAs like Doma and Keana. Extension services should provide training on cultivation techniques to maximize yields under flood conditions. This approach can reduce losses, as demonstrated in Southeast Asia, where flood-tolerant rice varieties have stabilized production in flood-prone regions .

5.2.2. Investment in Drainage and Flood Control Infrastructure

The concentration of flood impacts in low-lying areas along the Benue River underscores the need for improved drainage systems. The study flood extent maps (2022: 97,495.71 hectares; 2024: 63,577.41 hectares) highlight persistent inundation in Doma, Keana, and Toto. Investments in constructing and maintaining drainage canals, levees, and embankments can mitigate flood risks. For example, reinforcing riverbanks in these LGAs could prevent overflow during peak rainy seasons (July-September). The state government should collaborate with federal agencies like the National Emergency Management Agency (NEMA) to secure funding for such projects. Additionally, regular maintenance of existing drainage systems in rural areas, often neglected, is critical to ensure functionality, as poor drainage exacerbated the 2022 floods.

5.2.3. Development of Early Warning Systems

The integration of geospatial data with meteorological forecasts can enhance flood preparedness. The study use of Sentinel-1 SAR imagery for near-real-time flood mapping demonstrates the potential of geospatial technologies .The Nigerian Meteorological Agency (NiMet) should establish early warning systems that combine satellite-derived flood predictions with rainfall forecasts, delivering alerts to farmers via mobile platforms. Community-based dissemination channels, such as radio and local leaders, can ensure accessibility in rural areas. Such systems have proven effective in reducing agricultural losses in other flood-prone regions . Pilot projects in Doma and Keana could test the feasibility of these systems, with scaling potential across all 13 LGAs.

5.2.4. Land Use Planning and Zoning Regulations

The spatial distribution of flood impacts, concentrated in floodplains, calls for stricter land use planning. The study LULC analysis revealed that agricultural lands in flood-prone areas are highly vulnerable. The Nasarawa State government should enforce zoning regulations to restrict agricultural expansion into floodplains, promoting upland farming in LGAs like Lafia and Akwanga, where non-flooded agricultural areas were identified. Geospatial data from this study can guide the delineation of high-risk zones, ensuring that new farms and settlements avoid vulnerable areas. Incentives, such as subsidies for upland farming, could encourage compliance, reducing exposure to flood risks.

5.2.5. Strengthening Agricultural Insurance and Compensation Schemes

The economic impact of the 2022 floods (over N10 billion) and the 2024 floods highlights the need for robust agricultural insurance schemes. The Nigerian Agricultural Insurance Corporation (NAIC) should expand coverage to smallholder farmers, offering affordable premiums tailored to flood-prone regions. The study flood impact maps (Figures 710) provide precise data to support compensation claims, ensuring that affected farmers in Doma, Keana, and Toto receive timely support. NGOs can facilitate awareness campaigns to educate farmers on insurance benefits, addressing the low uptake noted in the literature .

5.2.6. Capacity Building and Community Engagement

Training programs for farmers on flood-resilient practices, such as raised-bed farming and crop diversification, can enhance adaptive capacity. The NSADP should organize workshops in collaboration with NGOs, focusing on communities in flood-prone LGAs. Community engagement is crucial, as women and marginalized groups face disproportionate impacts from flooding. Participatory mapping, using geospatial tools, can involve farmers in identifying vulnerable areas and planning mitigation measures, fostering ownership and resilience [Reforestation and Ecosystem Restoration]

The decline in vegetation cover (5.55% to 3.92% in flooded areas) reduces natural flood buffering . Reforestation initiatives along the Benue River can stabilize soils and reduce runoff. The state

government should partner with environmental NGOs to plant native species in degraded areas, particularly in Doma and Awe. Agroforestry, integrating trees with crops, can provide additional income sources while enhancing ecosystem services, as demonstrated in similar tropical regions .

5.2.7. Integration of Geospatial Technologies in Policy Frameworks

The study success in using Sentinel-1 and Sentinel-2 imagery (89.5% accuracy, Kappa 0.87) underscores the value of geospatial technologies for disaster management. The Nasarawa State government should institutionalize Remote Sensing (RS) and Geographic Information Systems (GIS) in disaster response and agricultural planning. Establishing a state-level geospatial unit, equipped with GEE and ArcGIS-Pro, can ensure continuous monitoring of flood risks and agricultural health. Training for government staff and researchers can build capacity, making Nasarawa a model for other Nigerian states.

5.3. Conclusion

5.3.1. Summary of Key Findings

This study assessed the impact of the 2022 and 2024 flood events on agriculture in Nasarawa State using RS and GIS techniques, achieving its aim and objectives. The key findings are:

1. **Flood Extent:** The 2022 floods inundated 97,495.71 hectares, reduced to 63,577.41 hectares in 2024, a 34.8% decrease, primarily affecting Doma, Keana, and Toto.
2. **LULC Changes:** Agricultural land was the most affected class (56.91% in 2022, 58.65% in 2024), with declines in vegetation and bareland indicating environmental shifts.
3. **Agricultural Impact:** Losses were 163,585.71 hectares (7.61%) in 2022 and 33,982.92 hectares (1.67%) in 2024, highlighting persistent vulnerability in floodplains.
4. **Geospatial Efficacy:** Sentinel-1 and Sentinel-2 imagery provided accurate, scalable insights, validated by high classification accuracy and alignment with field reports.

These findings confirm the severe threat of flooding to Nasarawa's agricultural sector, aligning with global trends where floods rank as a major hazard to agriculture . The geospatial approach

provided a cost-effective, scalable alternative to traditional surveys, addressing the research questions on flood extent, LULC dynamics, and agricultural impacts.

5.4. Contributions to Knowledge

The study contributes to the scientific understanding of flood dynamics in Nasarawa State by providing precise, spatially explicit data on flood extent and agricultural losses. It builds on previous works by offering a comparative analysis of two major flood events, highlighting temporal changes and persistent vulnerabilities. The use of open-access Sentinel data and GEE ensures reproducibility, making the methodology a replicable model for other flood-prone regions in Nigeria. For policymakers, the study provides evidence-based data to prioritize resources and design interventions, fulfilling its significance for government, NGOs, and farmers.

5.5. Limitations

Despite its strengths, the study has limitations. The reliance on satellite imagery, while effective, may miss micro-scale impacts detectable only through ground surveys. The temporal scope (2022 and 2024) limits long-term trend analysis, and the lack of real-time field data constrained validation in some LGAs. Additionally, socio-economic impacts, such as displacement and market disruptions, were inferred from secondary sources, not primary data. Future studies could integrate field surveys and socio-economic assessments for a holistic perspective.

5.6. Future Research Directions

Future research should:

- Conduct multi-year analyses to identify long-term flood trends and climate change impacts.
- Integrate socio-economic surveys to quantify impacts on livelihoods and food security.
- Explore machine learning for predictive flood modeling, enhancing early warning systems.

- Assess the effectiveness of proposed interventions, such as flood-tolerant crops and drainage systems, through pilot projects.

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